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Expected idiosyncratic volatility[☆]

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ABSTRACT

We use close to 80 million daily returns for more than 19,000 CRSP listed firms to establish the best forecasting model for realized idiosyncratic variances. Comparing forecasts from multiple models, we find that the popular martingale model performs worst. Using the root-mean-squared-error (RMSE) to judge model performance, ARMA(1,1) models perform the best for about 46% of the firms in out-of-sample tests. The ARMA(1,1) model delivers an average RMSE that is statistically significantly lower than all alternative models, and also performs well when not the very best. Its forecasts reverse large, unexpected shocks to realized variances. When using this model to revisit the relation between idiosyncratic risk and returns (the IVOL puzzle), we fail to find a significant relation. The IVOL puzzle is closely connected to a very small set of observations where the martingale forecast over-predicts the future realized variance. These extreme observations are correlated with well-known firm characteristics associated with the IVOL puzzle such as poor liquidity as measured by high bid-ask spreads and the "MAX" effect.

1. Introduction

Quantifying the expected variance of an individual stock position requires estimates of systematic exposure, the conditional variances of systematic variances (e.g. the market variance in a market model), and the expected idiosyncratic variance. There is a very extensive literature on the first two components,[1] but the literature on quantifying the expected idiosyncratic variance is scant. This is surprising as it is well-known that the idiosyncratic variance component, more often than not, dominates the systematic component of individual stock return variances.

In this paper, we build on the state-of-the-art literature on volatility forecasting to run a horse race between a variety of models to measure expected idiosyncratic risk. Our base set of models include the popular martingale model (see Ang et al., 2006), a variant of Corsi's (2009) heterogeneous autoregressive ("HAR") model, a non-linear

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[1] For example, the literature on predicting market variances is too vast to survey, but we use some of its state-of-the-art models, like the Corsi (2009) model, below.

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autoregressive model and an ARMA(1,1) model. To accommodate potential specification errors for the systematic component of returns and/or market wide variance components, we also consider the last three models with the market variance as an additional independent variable. Our models aim to forecast future monthly idiosyncratic realized variances using daily data for individual firm returns going back to 1926. Using mainly the Root Mean Squared (Forecast) Error, henceforth RMSE, we find that an ARMA(1,1) model for realized variances is the best model in terms of out-of-sample performance. It generates the best out-of-sample forecasts for more than 34% of firms. The ARMA model and the ARMA model with the market variance together generate the best forecasts for over 46% of firms. The ARMA(1,1) model also performs relatively well when not the very best, ranking among the top 3 models for more than 72% of firms. Furthermore, the median ratio between the RMSE of the ARMA model to that of the best model is 1.0106, meaning that the ARMA forecast is only 1.06% worse than the forecast of the best model for the median firm. More telling, the 75% percentile of this ratio is 1.0549 implying that the ARMA forecasts are very good even when not the very best. Finally, it delivers statistically significantly lower average RMSE than all alternative models. Economically, the ARMA model features an autoregressive coefficient much higher than an AR(1) model, but lower, of course, than the martingale model. However, through a large moving average coefficient its forecasts reverse large, unexpected shocks to realized variances.

The outperformance of the ARMA(1,1) should not come as a surprise given the literature on high frequency volatility forecasting (see e.g. Barndorff-Nielsen and Shephard, 2002). When the integrated variance follows a simple autoregressive process, the realized variance naturally follows an ARMA(1,1) process.[2] This literature also suggests various extensions of our base models, including higher order ARMA models (see e.g. Andreou and Ghysels, 2002), models embedding quarticity (see e.g. Bollerslev et al., 2016), and MIDAS models (see Ghysels et al., 2007). However, even compared to these more sophisticated models, the ARMA(1,1) model remains the best performing model. We perform a host of robustness checks to the measurement of idiosyncratic realized variances (IVAR) underlying our main results, and the results remain unchanged.

Our results offer a useful perspective on the huge literature on the (negative) pricing of idiosyncratic risk in the cross-section of expected returns (see Ang et al. 2006). We find that while expected idiosyncratic risk derived from the martingale model is negatively (and significantly) related to returns (as shown in prior literature), none of the other forecasts, including those of the best ARMA(1,1) model, are significantly related to returns. Our failure to reject the null of no relation continues to hold when generating the forecasts from the "best" model for each firm (judged by the lowest out-of-sample RMSE for each firm) and when back testing models each month and choosing the forecast from the model with the best past performance. Digging deeper, we document that the negative relationship between idiosyncratic risk and future realized returns using the martingale model is very fragile, and disappears when a limited number of firm-month observations are removed. Specifically, removing just 0.4% of the observations each month with the highest realized idiosyncratic variance forecast errors, measured as forecast minus realization, suffices to render the relationship insignificant. The relationship is also rendered insignificant in the univariate (multivariate) regression models when 5% (7%) of firm months with the highest martingale - ARMA(1,1) forecast differences are removed. Hence, the IVOL puzzle is driven by observations where the martingale forecast is particularly poor, and specifically when past IVAR shocks (i.e. high IVAR in time *t-1*) revert in time *t*, which the martingale fails to account for. Importantly, under identical sample variations for the ARMA(1,1) model, the inference regarding the relationship between idiosyncratic variances and returns does not change.

We then link these extreme forecast errors to standard controls used in IVOL-puzzle regressions and the explanatory variables explored in Hou and Loh (2016), using a variety of econometric techniques. We conjecture that firm characteristics associated with large martingale forecast errors should also help explain the IVOL puzzle. We find a strong relationship with bid-ask spreads, size indicators, measures of illiquidity, and past returns (the reversal effect). The variable most dramatically related to extreme forecast errors is the MAX variable (the maximum daily return over the past month) from Bali et al. (2011), which seems a good proxy for firm months in which the realized idiosyncratic variance is excessively and temporarily high.

The ARMA(1,1) forecasts differ drastically from the martingale model, as the former model allows temporary shocks to IVAR to revert, and does therefore not generate many of the poor forecasts (outliers with large forecast errors) that drive the negative relation between martingale IVOL and returns. This helps explain the difference in the relation between the forecasts from this model and returns. We perform the same analysis using the difference between the martingale and ARMA(1,1) forecasts and, indeed, the same firm characteristics that explain the extreme martingale forecast errors, are associated with large differences in the forecasts of the two models.

We further characterize what firm properties cause variation in expected idiosyncratic volatility. Focusing on the forecasts from the ARMA (1,1) model, we find that expected idiosyncratic volatility is increasing in beta and turnover while decreasing in firm size and book to market, with the size and turnover effects economically largest. Finally, we investigate the evolution of aggregate expected idiosyncratic volatility over time. It appears overall stationary but shows occasional extreme peaks, such as during the Great Depression, the 1970s' stagflation, the Tech boom and ensuing bear market, the Great Financial Recession, and the Covid shock.

Our paper provides useful inputs to several literatures. First, the literature on the dynamic properties of expected idiosyncratic volatility should be helpful in risk management, asset allocation and option pricing. In asset management, idiosyncratic volatility is a key determinant of the optimal weight assigned to mis-priced securities or arbitrage opportunities (see Pontiff, 2006; Treynor and Black, 1973). Many individual option pricing models (see e.g., Bakshi et al., 2021) start from models built for index options. Our results suggest that such models may miss an important dynamic component of idiosyncratic risk.[3]

Secondly, a rapidly growing macroeconomic literature studies the effect of uncertainty shocks on real economic activity and business cycles, and documents that heightened uncertainty can entail economic slowdowns through delayed firm investments, or increased precautionary savings by households. While macroeconomists often employ stock market data to measure uncertainty, they take various short-cuts by using aggregate market volatility or measures of cross-sectional return dispersion (see e.g., Bloom (2009) and Christiano et al. (2014)). However, the relevant economic models seem to call for a measure of *idiosyncratic* variance, reflecting non-systematic volatility, and, better still, a measure of firm-specific productivity or output uncertainty. Our measures may therefore provide useful input to this macroeconomic literature.[4]

Finally, our finding that expected IVOL is not priced when estimated using reasonable models should be comforting to proponents of efficient

[2] We thank the referee for pointing us in this direction.

[3] Bakshi et al. (2003) show, inter alia, that individual equity options reveal less negative skewness than market options. We estimate our models using projections or quasi maximum likelihood and do not take a stand on conditional distributions, which are very important in asset pricing. Our results suggest that the data generating process for individual stock variances should feature a shock dampening volatility effect, as implied by the ARMA(1,1) model.

[4] Kozeniauskas et al. (2018) distinguish between uncertainty shocks derived from both micro and macro data, but also show they are substantially correlated.

markets and traditional asset pricing theory. In addition, the findings that the negative relation between martingale IVOL forecasts and returns is driven by very few observations with particularly poor forecasts, and that explanations of the puzzle are all related to the incidence of these poor forecasts should be helpful in resolving the "IVOL Puzzle".

The remainder of the paper is organized as follows. The next section (2) describes the data. Section 3 describes the construction of our expected realized variance proxies using 7 different base models and a variety of more sophisticated models. Section 4 analyzes the forecast performance; Section 5 contrasts the pricing of idiosyncratic risk under different volatility models and analyzes why the martingale model uniquely delivers a negative relation with returns. Section 6 analyzes the properties of expected idiosyncratic volatility using the best (ARMA) model. The conclusion is presented in Section 7.

2. Data and variable descriptions

We obtain daily holding period returns inclusive of dividends, prices, shares outstanding, and volume data for equities traded on NYSE/AMEX, and NASDAQ from January of 1926 to December 2022 from the Center for Research in Security Prices (CRSP). This provides us with a total of 78,304,731 daily (3,715,261 monthly) firm returns covering 26,493 firms. The number of firms varies between 500 (1926:1) and 7,519 (1997:12) over the 1,164 available months. We obtain accounting data from Compustat's annual fundamentals file. The Fama and French (1993) factors and the risk-free rate (RF) are obtained through Wharton Research Data Services (WRDS). Our tests on the relative performance of different models (Tables 1-4) are based on data over the entire 1926–2022 sample. The remaining tables have a later start date based on data availability from Compustat as described below.

To characterize the results and conduct tests for whether idiosyncratic risk is priced, we collect data on firm characteristics. Beta, size, and market-to-book ratios are defined as in Fama and French (1992) and updated annually while turnover and the coefficient of variation of turnover are constructed following Chordia et al. (2001) and updated monthly. Turnover (TURN) is calculated as the average share turnover (monthly volume divided by the numbers of shares outstanding) in the past 36 months (a minimum of 24 months of data is required). We also calculate the coefficient of variation of the turnover over the same time period (CVTURN). Momentum (MOM) is calculated as the mean monthly holding period return between month t-7 to t-2, and Reversals (LAGRET) represents last month's return. Momentum and Reversals are calculated based on monthly data and updated monthly. All cross-sectional variables, except for beta, are calculated ex-ante.[5]

To explore how martingale IVOL forecasts may contribute to the IVOL puzzle, we collect data on common explanatory variables used in this literature (see e.g. Hou and Loh, 2016). Skewness (SKEW) is calculated using daily returns in the previous month while Co-skewness (COSKEW) is measured as the regression coefficient of squared daily stock returns on market returns in the previous month. We require 15 return observations within a month for both SKEW and COSKEW. We obtain expected idiosyncratic skewness (EXPIDIOSKEW) from Boyer et al. (2010).[6] We use several alternative measures of market frictions. Amihud Illiquidity (AMILL) is calculated following Amihud (2002) as the past 12-month average of daily returns divided by turnover while bid-ask spread (BIDASK) is calculated following Amihud and Mendelsohn (1986) as the effective bid-ask spread based on Corwin and Schultz (2012) scaled by the stock price. Both measures are downloaded from the Open Source Asset Pricing website[7] and these values are lagged to ensure that we do not introduce a look-ahead bias. Finally, Zero return (ZERORET) is measured following Han and Lesmond (2011) as the fraction of trading days with a zero return in the previous month. We also use two alternative size measures (both dummies). One is for firms that have a price below $5 at the end of the prior month (LOW_PRC), and another is for firms with a market capitalization below the 5th NYSE percentile in the prior month (LOW_MC). Finally, we measure MAX as the highest daily return in the prior month.

Although we use all available data to create our variables, many of our tests start in 1963 due to limitations in accounting data prior to that time. For our asset pricing tests, we follow prior literature on investigating whether idiosyncratic risk is related to returns in the cross-section and winsorize all continuous variables (except for returns[8] and BETA) monthly at the 0.5% and 99.5% levels.

3. Variance forecasting models

To compute the best possible estimate of expected idiosyncratic variances, we rely on the state-of-the-art volatility forecasting literature, which now almost exclusively uses realized variance models. Andersen et al. (2003) show that, under mild conditions, the expected conditional realized variance converges to the conditional return variance, where the realized variance is the quadratic variation of high frequency returns. Therefore, our criterion to define forecasting performance is the ability of different models to predict future realized idiosyncratic variances.[9] In Section 3.1, we outline the computation of idiosyncratic variances; in Section 3.2, we first discuss why theoretically, an ARMA(1, 1) model would be a natural forecasting model for realized variances, whereas the martingale model is mis-specified. We then present the 7 base models we examine. In Section 3.3, we discuss and justify various extensions of the base models. These include higher order ARMA(p,q) processes, models embedding quarticity (following Bollerslev et al., 2016), MIDAS models (Ghysels et al., 2019), and the monthly EGARCH Model (Nelson, 1991)

3.1. Calculating idiosyncratic realized variance

The extensive literature on volatility forecasting focuses mostly on index returns, where the consensus currently has converged to using 5-minute returns (see Andersen et al., 2001, and many others) as an adequate trade-off between microstructure noise and the continuous time limit. With microstructure noise (see e.g., Chordia and Subrahmanyam, 2004; Asparouhova et al., 2010; and Han and Lesmond, 2011 among others) more important for most individual firms than for market returns, sparser sampling might be preferred (see Bandi and Russell, 2006). Although an active econometric literature has proposed several new realized variance estimators to reduce the effects of microstructure noise (see e.g. Jacod et al. (2009)), we are forced to measure idiosyncratic shocks at the daily frequency because high frequency data for (a subset of) individual stocks is only available since the early nineties.

Let's define the idiosyncratic return shock for an individual firm i, observed at day d in month t, as $\varepsilon_{i,d,t}$. We define the monthly idiosyncratic realized variance ($IVAR_{i,t}$) as the sum of the squared daily idiosyncratic return shocks ($\varepsilon_{i,d,t}$) within month t. For our main models, we use $\varepsilon_{i,d,t}$ from the Fama and French (1993) model estimated within

[5] Fama and French (1992) argue the forward-looking BETA proxy is more precise than using BETA estimates at the firm level. While this variable is used in analyzing our results, as well as the asset pricing tests, it is not used to compute idiosyncratic risk and does not feature in our model selection exercises.

[6] We thank Prof. Brian Boyer for making the data available for download at https://boyer.byu.edu/skewness-data.

[7] https://www.openassetpricing.com/.

[8] Following prior literature, we set monthly returns over 300% to missing for our asset pricing tests.

[9] While a few articles forecast volatility (the square root of the variance), volatility is not a proper moment and constitutes a non-linear transformation of the variance.

month t.

$$RET_{i,d,t} - RF_{d,t} = a_{i,t} + b_{i,t}(MKT_{d,t} - RF_{d,t}) + s_{i,t}\text{SMB}_{d,t} + h_{i,t}\text{HML}_{d,t} + \varepsilon_{i,d,t}$$
$$IVAR_{i,t} = \sum_{d=1}^{N_t} \varepsilon_{i,d,t}^2 \tag{1}$$

where $RET_{i,d,t}$ is the stock return for firm i on day d in month t, $RF_{d,t}$ is the daily equivalent of the one-month Treasury bill rate from Ibbotson Associates on day d in month t obtained from Professor Ken French (through WRDS), and MKT, SMB and HML are the market, size and value factors. N_t is the maximum number of trading days within the month.

Our definition of idiosyncratic shocks closely follows the finance literature, but we also consider several additional alternatives for robustness (See Section 4.4 for the results). First, we use the CAPM in month t instead of the Fama-French three factor model. Second, we estimate the parameters of the Fama-French three factor model using daily data in the prior 3 months (t-2 to t), to estimate $\varepsilon_{i,d,t}$ in month t. Third, we adjust $IVAR$ for autocorrelation following French et al. (1987) by adding 2 times the sum of daily cross-products of adjacent returns. Specifically,

$$IVAR_{i,t}^{FSS} = \sum_{d=1}^{N_t} \varepsilon_{i,d,t}^2 + 2\sum_{d=1}^{N_t-1} \varepsilon_{i,d,t}\varepsilon_{i,d+1,t} \tag{2}$$

where, as stated above, the subscript i is the firm, t is the month, d is the day within the month, and N_t is the maximum number of trading days within the month. Fourth, we restrict the sample to only include firms in the S&P500 in a given month, as this should ensure that the findings are not driven by illiquidity or micro-structure noise.

3.2. Predictive models for realized variances

Ang et al. (2006) find that lagged realized volatility is negatively related to returns (which is often termed the IVOL Puzzle), implicitly using a martingale model for realized variances. This is the most popular IVAR model and the first model we consider:

$$IVAR_{i,t} = IVAR_{i,t-1} + e_{i,t} \tag{3}$$

However, on theoretical grounds, this model is almost surely misspecified. Without getting into technical details, the realized variance literature shows that under certain conditions, the conditional return variance can be measured as the conditional expected value of integrated variance, which is approximated by the realized variance. Let's define integrated variance as IV_t, and realized variance (measured using a particular sampling interval) as RV_t, then, $RV_t = IV_t + u_t$ where u_t is a zero mean noise term, reflecting microstructure noise and the fact RV_t uses a particular sampling frequency during the day (See Anderson et al., 2003; Barndorff-Nielsen and Shephard, 2002 for formal discussions). If we assume an AR(1) model for IV_t, and standard white noise for the error term u_t, the resulting time series model for RV_t is an ARMA(1,1) model. Thus, there are theoretical reasons to expect the ARMA(1,1) model to outperform the martingale model, and it is the second model we consider:

$$IVAR_{i,t} = a_i + b_i IVAR_{i,t-1} - \theta_i e_{i,t-1} + e_{i,t} \tag{4}$$

We estimate this model by Maximum Likelihood (MLE), assuming the initial residual is zero. Economically, when a large volatility spike occurs, it likely causes a large positive residual error; with a positive θ coefficient, the model can then counteract the high volatility prediction of the autoregressive component.

Econometrically, there is an alternative way to accommodate relatively quick changes in the dependence on past variances, namely time-varying autoregressive parameters. It is conceivable that the persistence of volatility is different in normal times than in high volatility times. For example, imagine a firm experiencing a large earnings shock, which causes its volatility to spike. Presumably, such large volatility spike should not persist into the next month. To accommodate such variation in persistence, our third model is a non-linear AR(1) model, where the persistence coefficient is a function of the past variance ("ARNL"):

$$IVAR_{i,t} = a_i + \theta_i {}^* L(b_i {}^* IVAR_{i,t-1}) {}^* IVAR_{i,t-1} + e_{i,t} \tag{5}$$

where $L(.)$ is the logistic function, i.e., $L(x) = exp(x)/[1 + exp(x)]$. We expect the b-coefficient to be negative. Bekaert et al. (2024) find evidence for such non-linearity in US, Euro area and Japanese market variances.[10] The model is estimated by non-linear least squares.

The fourth model is a variant of the Corsi (2009) "HAR" model, perhaps the most popular model in the extensive realized variance forecasting literature at the market level:

$$IVAR_{i,t} = a_i + b_i IVAR_{i,t-1} + \theta_i IVAR_{i,t-1,w} + c_i IVAR_{i,t-1,d} + e_{i,t} \tag{6}$$

where $IVAR_{i,t-1,w}$ is the sum of squared residuals in the last week (w) of the prior month (t-1) for firm i and $IVAR_{i,t-1,d}$ is the squared residual on the last trading day (d) in the prior month (t-1) for firm i. Thus, more recent information can differentially affect the conditional variance for the next month. The HAR model was initially developed for daily realized variance forecasting and the current formulation may not be optimal for monthly forecasting. However, the results in Bekaert and Hoerova (2014) for stock return variances at the market level, and for various asset classes in Ghysels et al. (2019) show that it performs reasonably well for monthly horizons. In essence, the HAR model is a MIDAS regression with step functions, see e.g. Ghysels et al. (2007) and the relevant discussion in Corsi (2009). If optimal at the daily level, the monthly forecast could also be viewed as a special case of a MIDAS forecast, putting different weights on the various daily realized variances within the month. We consider a MIDAS framework directly in Section 3.3.

In addition to the previous 4 models, we also estimate each of the last 3 models including the sum of squared daily returns of the market during the prior month (i.e., the market realized variance) as an independent variable. While we have removed systematic components in returns, there is evidence of common factors in idiosyncratic firm variances (see Herskovic et al., 2016, for evidence in the US, and Bekaert et al., 2024, for evidence in 21 countries.) Allowing dependence on the market variance is a simple way to potentially capture such factors, which may improve the forecasting power of the model.[11]

3.3. Additional predictive models for realized variances

While the bulk of results focuses on the base models introduced in Section 3.2, we also estimate a wide set of alternative models. These models fall into four different categories: generalizations of the ARMA (1,1) model, alternative time-varying parameter models embedding quarticity, MIDAS models which generalize the HAR model, and the EGARCH model.

3.3.1. Additional ARMA models

While theory suggests that realized variances follow an ARMA(1,1) model, this assumes that the integrated variance is an AR(1) process. Chernov et al. (2003), among others, find that continuous time volatility models are best described by two-factor models, which would suggest an ARMA(2,2) model for realized variances. Component models of

[10] The model is also related to the quarticity correction proposed in Bollerslev et al. (2016), which models the persistence coefficient as a function of the degree of uncertainty with which the realized variance is measured, see Section 3.3.

[11] Bartram et al. (2016) and Bekaert et al. (2012) find a strong link between the aggregate idiosyncratic variance and the conditional market variance.

volatility (e.g., the permanent and transitory components in Engle and Lee, 1999; or bad and good volatility components in Bekaert et al., 2015) also suggest richer ARMA processes. In fact, Barndorff-Nielsen and Shephard (2002) show theoretically that integrated and realized variances are ARMA(p,p) processes when the spot variance is a linear combination of p independent continuous time autoregressive processes.

An alternative literature, focusing on improving the measurement of realized variances, suggests using smoothing by incorporating lagged realized variances (see e.g. Ghysels et al., 2023 and Andreou and Ghysels, 2002). The Andreou-Ghysels paper suggests that for the S&P 500 index, using up to two lags improves measurement of monthly realized variances. Such measurement schemes would also suggest more complicated ARMA models for the realized variance. We therefore consider an alternative forecasting model, where we estimate all (12) ARMA(p,q) models, with p in the (1,2,3) set and q in the (0,1,2,3) set. For each firm, in each month (t), we then use the forecast for month ($t+1$) from the model with the lowest Akaike information criterion (1974, AIC henceforth).

While theory may favor more complicated models, it is well known that in out-of-sample forecasting, parsimonious models do well. We therefore also consider a simple autoregressive AR(1) model for realized variances:

$$IVAR_{i,t} = a_i + b_i IVAR_{i,t-1} + e_{i,t} \tag{7}$$

3.3.2. Quarticity models

Ghysels et al. (2023) derive the optimal weight on current and past realized variances in the measurement of integrated variance, suggesting the weight on current realized variance should depend on the noise with which it is measured. This noise is proportional to the quarticity of realized variances, which depends on returns to the fourth power (see e.g. Barndorff-Nielsen and Shephard, 2002). Bollerslev et al. (2016) focus directly on forecasting future realized variances and suggest estimating an AR(1) model with the AR(1) coefficient interacted with realized quarticity, or an HAR model with some or all of the coefficients interacted with the relevant quarticity measure. Note that the intuition here is quite similar to the non-linear autoregressive model (ARNL) in the base set of models. We estimate a number of such models.

3.3.2.1. Autoregressive quarticity model

$$IVAR_{i,t} = \alpha_{i,0} + \left(\alpha_{i,1} + \alpha_{i,1Q}\sqrt{RQ_{i,t-1}}\right)IVAR_{i,t-1} + e_{i,t} \tag{8}$$

where RQ stands for realized quarticity, and RQ is generally estimated as: $RQ_{t,d} = (M/3)\sum_{k=1}^{M} r_{t,k}^4$, where $r_{t,k}$ is the high frequency (e.g. 5 min) return, and M is the number of high frequency returns used per day, where we dropped the stock subscript. Of course, in our set up with daily returns, $M = 1$, so that daily quarticity is the daily idiosyncratic return to the 4th power and monthly quarticity becomes $RQ_t = (1/N_t)(1/3)\sum_{k=1}^{N_t} r_{t,k}^4$ where $r_{t,k}$ are now idiosyncratic daily returns within month t and N_t is the number of trading days within month t. The prediction is that α_{1Q} is negative; when the noise in measuring RV is high, the persistence of realized variances decreases.

3.3.2.2. HAR quarticity model. Here all coefficients in the HAR model interact with the relevant quarticity measure:

$$\begin{aligned} IVAR_{i,t} = {} & \alpha_{i,0} + \left(\alpha_{i,1} + \alpha_{i,1Q}\sqrt{RQ_{i,t-1}}\right)IVAR_{i,t-1} \\ & + \left(\beta_{i,1} + \beta_{i,1Q}\sqrt{RQ_{i,t-1,w}}\right)IVAR_{i,t-1,w} \\ & + \left(\gamma_{i,1} + \gamma_{i,1Q}\sqrt{RQ_{i,t-1,d}}\right)IVAR_{i,t-1,d} + e_{i,t} \end{aligned} \tag{9}$$

where the adjustment for the weekly (5 days) and one day quarticity measures sums the daily returns to the 4th power over the recent 5 (1) days and divides by 5 times 3 for the weekly estimate or by 1 times 3 for the daily estimate.

3.3.2.3. ARMA(1,1) quarticity model. In this model, both the AR and MA coefficients are a linear function of the quarticity measure.

$$\begin{aligned} IVAR_{i,t} = {} & \alpha_{i,0} + \left(\beta_{i,1} + \beta_{i,1Q}\sqrt{RQ_{i,t-1}}\right)IVAR_{i,t-1} \\ & + \left(\theta_{i,1} + \theta_{i,1Q}\sqrt{RQ_{i,t-1}}\right)e_{i,t-1} + e_{i,t} \end{aligned} \tag{10}$$

For the above quarticity models, we set the starting values based on the median values from the "base" models, and we set the quarticity coefficients to 0. All quarticity models are estimated using PROC MODEL in SAS.

3.3.2.4. MIDAS quarticity model. We defer the discussion of this model to the next section.

In all these models, we follow Bollerslev et al. (2016) and do not use RQ itself in the estimation but RQ minus its sample mean so that $\beta_{i,1}$, for example, is the unconditional autoregressive estimate. In our out-of-sample analysis, this averaging is done for each rolling sample over which we conduct the estimation.

3.3.3. MIDAS models

As mentioned before, the HAR model is a special case of a MIDAS model. We therefore also consider MIDAS models explicitly: that is, we model the forecast as a function of all $IVAR_{i,t,d}$ in the prior 22 days with a flexible weight function. We parametrize the weights with an (exponential) Almon lag specification as in Ghysels et al. (2019). Obviously, the monthly implied model of the daily HAR model would be a special case of this model. Specifically, this model can be written as follows:

$$IVAR_{i,t} = \alpha_i + \phi_i \sum_{j=0}^{K}\left(w_{i,j}\, IVAR_{i,t-1,d}\right) + e_{i,t} \tag{11}$$

where the t subscript is a month subscript and d indicates the day within the month, with $j = 0$ representing the last day in the month (the one that is nearest), and K the first day. We set $K = 22$. The weight function depends on two parameters, with the weights adding to one and always positive. Dropping the stock subscript i, the weight for day j is:

$$w_j = exp\left(\theta_1 j + \theta_2 j^2\right) \Big/ \sum_{j=0}^{K} exp\left(\theta_1 j + \theta_2 j^2\right)$$

Note that this specification has three parameters, two determining the weight function and one "autoregressive" parameter. This weight function can fit almost any pattern, for example, if $\theta_2 < 0$ weights decline to zero eventually. We estimate the model by non-linear least squares, just as we did for the nonlinear AR model. We use agnostic starting values, setting $\theta_1 = \theta_2 = 0$, which produces equal $1/K$ weights. For ϕ, we try different starting values in the set (11,9,7,5,3) and use the one that produces the smallest objective function value to start the iterative process. These starting values correspond to a wide, reasonable range of AR(1) coefficients. Note that for weights equal to $1/K$, the usual autoregressive coefficient would correspond to ϕ/K.

Running this non-linear estimation for our large set of stocks over a long historical period is not trivial, and we did experience convergence issues in about 8% of the cases. To improve the estimation performance, we also estimate simpler versions of the MIDAS model, namely a first order and second order Taylor approximation to the actual model. Internet Appendix 3 describes the derivation of these models. These models typically perform about as well as the original model and are more stable. Therefore, our estimation approach uses the quadratic model when the full model fails to converge, and the linear model when the quadratic model also fails to converge.

We also estimate the quarticity version of this model:

$$IVAR_{i,t} = \alpha_i + \left(\phi_{i,0} + \phi_{i,1}\sqrt{RQ_{i,t-1}}\right)\sum_{j=0}^{K}\left(w_{i,j}\, IVAR_{i,t-1,d}\right) + e_{i,t} \tag{12}$$

3.3.4. EGARCH models

Finally, we also consider Nelson's (1991) EGARCH model to predict future realized variances using monthly returns. In fact, this is one of the most popular models in the idiosyncratic volatility literature (See Fu, 2009; Fink et al., 2012; and Guo et al., 2014). However, as shown by Bergbrant and Kassa (2021) the model is a poor fit for most firms, and consistent with this we find that its performance is by far the worst from all the models we consider. Internet Appendix 3 describes how, consistent with the literature, we estimate this model and we later briefly further comment on its performance.

4. Forecast performance

Section 4.1 describes the practical setup of our forecasting exercise. Section 4. 2 discusses the out-of-sample results for the base set of models, whereas Section 4.3 discusses the relative performance of the additional models considered and Section 4.4 presents robustness tests

4.1. Forecasting set-up and in sample results

All our predictive models are estimated using monthly data. We require at least 36 monthly observations to include a firm in our sample. While we conduct a thorough in sample analysis, its results are mostly confirmed by the out-of-sample analysis. We therefore relegate a detailed discussion of the in-sample tests to the Internet Appendix (Internet Appendix 1).

For the out-of-sample exercise, we opt to use relatively long rolling samples of 10 years. This choice strikes a balance between having sufficient data to efficiently estimate model parameters and accommodating varying volatility dynamics over a firm's life cycle. This is feasible because our data go back to 1926 for many firms.[12] Concretely, we use rolling regressions, allowing up to 120 months (10 years) of data to estimate the models. We fit the models with data up until time *t*-1, and use the forecast for time *t* as our out-of-sample forecast. As indicated above, we require a minimum of 36 observations to generate the forecast for the following month. (Hence, for firms within their first 10 years of listing, the forecasts use an expanding window). In total, we estimate 17 out-of-sample models, and we only include a forecast for firm *i* at time *t*, when calculating statistics for each model, if we have forecasts from all our estimated models for the firm in that month. Our main performance criterion is the out-of-sample RMSE (the square root of the mean squared forecast errors).

With samples sometimes being relatively short, some models produce real outlier forecasts that any reasonable practitioner would reject. Although this does not impact the inferences regarding the best model, we winsorize all variance forecasts at 0 and 0.5 in our main tests. The 0.5 maximum value is close to the 99.5th percentile of realized variances in our full sample. (In robustness exercises, presented in Panels E and F of Appendix A, we redo the main tables with no winsorization and winsorization at the 2.5% and 97.5% levels and find qualitatively similar results.)

To provide a formal statistical test that any model indeed performs better in the cross-section of US stocks, we conduct three paired significance tests. Each test produces a test statistic for the null hypothesis that the mean (or median) of the RMSE is equal across two models (i.e. the difference between them is equal to 0) against the two-sided alternative that the mean or median is not equal. The first is the standard Student's *t*-test, which is appropriate if the data is from an approximately normal distribution. Our second and third tests conduct non-parametric tests of the differences in RMSEs between models, specifically the Sign Test and the Wilcoxon Signed Rank Tests.[13] Because these tests are quite standard, we relegate a brief description of them to Internet Appendix 1, where they are discussed together with the in-sample results.

4.2. Out-of-Sample performance for base models

We report the main out-of-sample performance results for the base models in Table 1, using the RMSE to measure forecast errors. The mean (median) RMSE is reported in the first (last) column. We produce average firm statistics in two ways: the first is to simply average the RMSE across firms and the second is to also weight the RMSE by the number of monthly observations available per firm. Although the weighted results show substantially lower average RMSEs, consistent with expectations that model performance improves with the number of observations used, the relative performance of the models are almost identical with the two methodologies, so we only report and discuss the simple arithmetic averages. The table shows that the martingale model is the worst model, but it is not too far off from the HAR model that uses the market variance. The best model is by far the simple ARMA(1,1) model both in terms of the mean and median RMSE.

In columns 2 and 3, we report the fraction of times the model is the best model (lowest RMSE) and the average rank. That is, for each firm we rank the performance of all models (with 1 being best and 7 being worst) and record the average rank. The ARMA(1,1) model is also best in terms of average rank (2.63) and is the best model in 34.3% of the cases. Together, the ARMA model and the ARMA model appended with market variance are the best for 46.5% of the cases. Despite featuring a higher average and median RMSE than all other models, the martingale model is the best model in 13.8% of the cases, and only two additional models other than the ARMA(1,1) model are best in more than 10% of the cases (the simple ARNL model and the ARMA model with the market variance). In terms of average rank, the second-best model is the ARNL model (3.38), followed closely by the ARMA model with the market

Table 1
Out-of-sample forecast performance for base models.

Model	Mean	Best	Rank	Imp. to Worst	Median
Martingale	0.1031	13.83%	5.32	29.41%	0.0324
ARMA(1,1)	0.0989	34.28%	2.63	81.80%	0.0275
ARNL	0.1005	17.35%	3.38	70.15%	0.0281
HAR	0.1021	8.18%	4.37	57.11%	0.0300
ARMA(1,1) w/ Mkt	0.1002	12.24%	3.45	72.15%	0.0287
ARNL w/ Mkt	0.1013	9.20%	3.95	63.82%	0.0292
HAR w/ Mkt	0.1030	4.78%	4.90	50.20%	0.0310

This table shows out-of-sample statistics for the 7 models described in the data section, where the statistics presented are based on firm averages. To calculate the out-of-sample statistics, each model is fit every month for each firm to generate a forecast for the following month, using up to 10 years of data (when available) prior to the time of the forecast. Before using these forecasts to calculate the out-of-sample statistic for each firm, all monthly forecasts are winsorized at 0 and 0.5. In addition to the mean statistic, the table also shows the proportion of firms for which a model generates the best forecasts, as well as the average rank of the model. We also report the "improvement to worst", defined as the average ratio of the difference between the worst RMSE and the model being examined relative to the RMSE difference between the worst and best. The final column reports the median RMSE. The sample period is from 1926 to 2022.

[12] This long-term perspective makes it impossible to consider the use of option implied volatilities, which are only available since 1996, and only for a small subset of our stock universe.

[13] Details about the estimation of the t-test, sign and Wilcoxon signed rank tests can be found here: Tests for Location : Base SAS(R) 9.4 Procedures Guide: Statistical Procedures, Third Edition.

variance (3.45). Taken together, these results clearly confirm the superiority of the ARMA(1,1) model with regard to forecasting idiosyncratic variances.

Because the average RMSEs appear close to one another, it is important to verify that the improvement in fit is economically meaningful and different across models. We therefore report the "Improvement to worst statistic," in column 4; that is, the average ratio of the difference between the worst RMSE and the RMSE of the model being examined, relative to the RMSE difference between the worst and the best model. If the model always has the lowest RMSE, the statistic is obviously one. The martingale model has a very low improvement to worst statistic of only 29.41%. The ARMA(1,1) model is by far the best, on average capturing 81.80% of the worst to best difference. The second to third best models are the ARMA model with the market variance, and the ARNL model, with ratios of around 70%.

The superiority of the ARMA(1,1) model is not driven by the specific time period used in this sample, as it is robust across different sub-periods. Internet Appendix 2 shows that it produces the best forecasts (when judging based on a combination of all criteria) in each decade of our study (from the 1930s to the 2020s).

These results also hold up in the formal paired statistical tests, reported in Table 2, with Panel A reporting the *t*-test, Panel B the Sign test and Panel C the Signed Rank test. Note that because we have so many firms, the tests are extremely powerful and almost invariably reject the null of equality with very small p-values.[14]

The ARMA(1,1) model produces significantly lower average RMSE

Table 2
Formal tests of Out-of-Sample (OOS) differences in RMSE.

Panel A: T-Test					w/ Mkt		
	Martingale	ARMA	ARNL	HAR	ARMA	ARNL	HAR
ARMA(1,1)	−32.88		−16.40	−28.99	−17.47	−22.33	−34.47
	(0.00)		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
ARNL	−16.18	16.40		−13.16	2.58	−13.63	−19.52
	(0.00)	(0.00)		(0.00)	(0.01)	(0.00)	(0.00)
HAR	−7.11	28.99	13.16		15.24	6.17	−14.85
	(0.00)	(0.00)	(0.00)		(0.00)	(0.00)	(0.00)
ARMA(1,1) w/ Mkt	−19.58	17.47	−2.58	−15.24		−10.58	−25.95
	(0.00)	(0.00)	(0.01)	(0.00)		(0.00)	(0.00)
ARNL w/ Mkt	−11.18	22.33	13.63	−6.17	10.58		−14.72
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)		(0.00)
HAR w/ Mkt	−0.72	34.47	19.52	14.85	25.95	14.72	
	(0.47)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	
Panel B: Sign Test					w/ Mkt		
	Martingale	ARMA	ARNL	HAR	ARMA	ARNL	HAR
ARMA(1,1)	−5932		−2719	−5136	−3927	−3525	−5401
	(0.00)		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
ARNL	−4729	2719		−3455	339	−2843	−4017
	(0.00)	(0.00)		(0.00)	(0.00)	(0.00)	(0.00)
HAR	−3470	5136	3455		2967	1695	−2636
	(0.00)	(0.00)	(0.00)		(0.00)	(0.00)	(0.00)
ARMA(1,1) w/ Mkt	−4836	3927	−339	−2967		−1858	−4528
	(0.00)	(0.00)	(0.00)	(0.00)		(0.00)	(0.00)
ARNL w/ Mkt	−3936	3525	2843	−1695	1858		−3573
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)		(0.00)
HAR w/ Mkt	−2622	5401	4017	2636	4528	3573	
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	
Panel C: Signed Rank Test					w/ Mkt		
	Martingale	ARMA	ARNL	HAR	ARMA	ARNL	HAR
ARMA(1,1)	−57.47		−30.00	−54.47	−41.40	−38.33	−57.21
	(0.00)		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
ARNL	−42.72	30.00		−34.95	3.55	−29.82	−41.19
	(0.00)	(0.00)		(0.00)	(0.00)	(0.00)	(0.00)
HAR	−28.43	54.47	34.95		31.28	16.98	−28.29
	(0.00)	(0.00)	(0.00)		(0.00)	(0.00)	(0.00)
ARMA(1,1) w/ Mkt	−45.88	41.40	−3.55	−31.28		−20.47	−48.01
	(0.00)	(0.00)	(0.00)	(0.00)		(0.00)	(0.00)
ARNL w/ Mkt	−34.69	38.33	29.82	−16.98	20.47		−35.73
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)		(0.00)
HAR w/ Mkt	−19.53	57.21	41.19	28.29	48.01	35.73	
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	

Panels A shows Student's *t*-test statistic and the associated *p*-value on the difference between the OOS RMSE (Panel A) for the model identified in column 1 minus that on the horizontal axis. For example, in the first cell is the *t*-stat for the difference in OOS RMSE betwee`n the ARMA(1,1) and Martingale model. Underneath the *t*-stat is the associated *p*-value. Panel B shows the Sign Test statistic and the associated *p*-value and Panel C shows the Signed Rank Test Statistics and the associated *p*-value. All Signed Rank Test statistics have been divided by 1 million to conserve space in the table.

[14] These tests assume that the firm values are uncorrelated across firms. Because the underlying observations represent a difference between test statistics (such as the RMSE) of two different models for a firm, it is difficult to imagine they would exhibit strong cross-firm correlation, and it is not clear such correlation would even be positive.

Table 3
Comparison of performance and forecasts for base models.

Panel A: Comparison to "best":	P10	P25	P50	P75	P90
Martingale	0.00%	6.08%	19.76%	30.86%	41.84%
ARMA(1,1)	0.00%	0.00%	1.06%	5.49%	17.63%
ARNL	0.00%	0.54%	3.43%	9.66%	26.07%
HAR	0.17%	2.32%	6.89%	16.09%	38.48%
ARMA(1,1) w/ Mkt	0.00%	0.65%	3.04%	9.70%	28.88%
ARNL w/ Mkt	0.04%	1.19%	4.80%	13.07%	36.01%
HAR w/ Mkt	0.59%	3.01%	8.60%	20.03%	49.91%

Panel B: Forecast Correlation					w/ Mkt		
	Martingale	ARMA(1,1)	ARNL	HAR	ARMA(1,1)	ARNL	HAR
Martingale	1						
ARMA(1,1)	0.82	1					
ARNL	0.77	0.87	1				
HAR	0.79	0.86	0.82	1			
ARMA(1,1) w/ Mkt	0.79	0.95	0.84	0.84	1		
ARNL w/ Mkt	0.76	0.85	0.95	0.80	0.87	1	
HAR w/ Mkt	0.77	0.84	0.80	0.95	0.87	0.84	1

Panel A shows the percentiles (10th, 25th, 50th, 75th, and 90th) of the distribution of a statistic measuring how much worse a given model is compared to the "best" model, namely the percentage increase in RMSE compared to the "best" model. Panel B shows the Pearson correlation coefficients between the forecasts produced by the different models.

compared to all other models; and the second-best model is the ARMA (1,1) model featuring the market variance, according to all three tests. The ARNL model is third best. It is striking that the results are very consistent across all three tests.

In addition, it turns out that the performance of the ARMA(1,1) model is stellar for the large majority of the firms. That is, even for the firms for which it is not the best model, its relative performance is still very good. Table 3 (Panel A) shows how each model fares compared to the "best" model. For each firm, we calculate how much larger (in percent) the RMSE is for Model i compared to the "best" model for that firm. We then present percentiles of the distribution across firms of this statistic. For example, for the martingale model, almost 10% (90th percentile) of the firms have a 42% worse performance than the best model. As is apparent, the ARMA(1,1) model works great in forecasting *IVAR* even when not the best. In fact, for 50% (75%) of firms, it generates forecasts with RMSEs that are less than 1.06% (5.49%) worse than the "best" model. The ARNL, ARNL plus market, and ARMA(1,1) plus market models also have median statistics less than 5%, with the ARMA (1,1) plus market model second-best at 3.04%. However, at the 90th percentile, the ARNL model is better than the ARMA(1,1) plus market model. The ARMA(1,1) model generates the lowest statistics across the whole distribution.

To gain some intuition on the differences between the forecasts generated by the various models, Panel B of Table 3 reports panel correlations between the forecasts of the 7 models. The forecasts of all models do show substantial correlation. Most of these correlations are between 0.75 and 0.95 with the martingale model generally producing forecasts that are the least correlated with the other models (correlations varying between 0.76 and 0.82).

Not surprisingly, the forecasts of the ARMA(1,1) model are most correlated with the forecasts from the ARMA model featuring the market variance (the correlation is 0.95). In fact, each model that includes the market variance has a correlation with its counterpart without the market variance that is 0.95.

4.3. Out-of-Sample performance for additional models

Here we cast a wider net and also discuss the performance of the additional models described in Section 3.3. The results are reported in several panels in Table 4. In Panel A, we compare the performance of more complicated ARMA models (where we select the best ARMA model), termed ARMA(AIC), and a simple AR(1) model with the four base models (martingale, ARMA(1,1), ARNL and HAR). We drop the models with the market variance as they are dominated by models without the market variance in out-of-sample exercises. Both new models are in fact quite competitive, with both beating the martingale and HAR models for most statistics (mean, median, rank, and improvement to worst). The simple AR(1) model is overall actually slightly better than the ARMA(AIC) model. It is also competitive with the ARNL model for most statistics, but the ARNL model is far superior in terms of the fraction of times it is the best model. The ARMA(1,1) model is by far the best model for all statistics.

Panel B examines whether embedding quarticity improves forecasting performance. We consider the AR(1), ARMA(1,1) and HAR models with their quarticity counterparts. It is very clear that incorporating quarticity in the models is not helpful and invariably increases the average and median RMSE and worsens the rank and "improvement to worst statistics". The only exception is that the AR(1) model with quarticity is best in 19.2% of the cases which is better than the simple AR (1) model. This is similar to the ARNL model being much better along this statistic than the AR(1) model in Panel A. Over all the statistics, the ARMA(1,1) model remains the best model.

In Panel C, we report the results for the MIDAS model together with the 4 non-market variance base models. The performance of the MIDAS model is very close to that of the HAR model, and only improves on it in terms of the fraction of times it is the best model (at 10.89% versus 9.36% for the HAR model). Again, the ARMA(1,1) model is by far the best model.

Finally, in Panel D, we report results for all 17 models we estimate, including the EGARCH model. Because of the terrible performance of the EGARCH model, the improvement to worst statistics go up considerably for most models, including the martingale model. The dominance of the ARMA(1,1) model is, if anything, even more glaring. It is the best model in 21.74% of the cases and its improvement to worst statistic is over 90%. The second-best model depends on the statistic considered: it is the AR(1) and ARMA(1,1) plus market variance models for the average RMSE, the martingale model for the best model fraction (at 9.24%), and the ARNL model for the average rank, the improvement to worst statistic and median RMSE.

We conclude that the ARMA(1,1) model is by far the best forecasting model for idiosyncratic variances.

Table 4
Out-of-sample forecast performance for additional models.

Panel A: Model	Mean	Best	Rank	Imp. to Worst	Median
Martingale	0.1031	14.28%	4.72	27.11%	0.0324
ARMA(1,1)	0.0989	36.27%	2.36	82.22%	0.0275
ARNL	0.1005	20.78%	3.16	69.60%	0.0281
HAR	0.1021	9.86%	4.09	55.71%	0.0300
ARMA(AIC)	0.1009	8.67%	3.51	66.25%	0.0285
AR(1)	0.1002	9.87%	3.16	69.92%	0.0284

Panel B: Model	Mean	Best	Rank	Imp. to Worst	Median
ARMA(1,1)	0.0989	41.87%	2.24	85.52%	0.0275
ARMA(1,1) Quart	0.1113	8.83%	4.69	32.67%	0.0403
HAR	0.1021	9.45%	3.68	66.90%	0.0300
HAR Quart	0.1088	6.83%	4.57	41.89%	0.0372
AR(1)	0.1002	13.61%	2.88	76.10%	0.0284
AR(1) Quart	0.1017	19.24%	2.94	75.13%	0.0294

Panel C: Model	Mean	Best	Rank	Imp. to Worst	Median
Martingale	0.1031	13.83%	3.89	29.08%	0.0324
ARMA(1,1)	0.0989	42.99%	2.04	82.35%	0.0275
ARNL	0.1005	22.81%	2.54	70.44%	0.0281
HAR	0.1021	9.36%	3.24	57.31%	0.0300
MIDAS	0.1024	10.89%	3.30	54.75%	0.0302

Panel D: Model	Mean	Best	Rank	Imp. to Worst	Median
Martingale	0.1031	9.24%	11.19	76.96%	0.0324
ARMA(1,1)	0.0989	21.74%	4.91	91.11%	0.0275
ARMA(AIC)	0.1009	4.46%	7.55	86.66%	0.0285
ARMA(1,1) Quart	0.1113	5.27%	12.05	63.64%	0.0403
ARNL	0.1005	7.39%	6.54	88.18%	0.0281
HAR	0.1021	3.34%	8.97	83.94%	0.0300
HAR Quart	0.1088	3.55%	11.67	69.96%	0.0372
AR(1)	0.1002	4.29%	6.76	87.78%	0.0284
AR(1) Quart	0.1017	7.44%	7.19	85.72%	0.0294
AR(1) w/ Mkt	0.1012	3.00%	8.02	85.48%	0.0295
ARNL w/ Mkt	0.1013	4.87%	7.75	86.12%	0.0292
ARMA(1,1) w/ Mkt	0.1002	7.22%	6.60	88.08%	0.0287
ARMA(AIC) w/ Mkt	0.1023	3.25%	9.15	83.50%	0.0299
HAR w/ Mkt	0.1030	2.44%	10.01	81.73%	0.0310
MIDAS	0.1024	5.36%	9.20	83.04%	0.0302
MIDAS Quart	0.1046	4.93%	9.86	78.83%	0.0326
EGARCH	0.1397	1.98%	15.58	16.09%	0.0718

This table shows out-of-sample statistics for subsets of the 17 models described in the data section, where the statistics presented are based on firm averages. To calculate the out-of-sample statistics, each model is fit every month for each firm to generate a forecast for the following month, using up to 10 years of data (when available) prior to the time of the forecast. Before using these forecasts to calculate the out-of-sample statistic for each firm, all monthly forecasts are winsorized at 0 and 0.5. In addition to the mean statistic, the table also shows the proportion of firms for which a model generates the best forecasts, as well as the average rank of the model. We also report the "improvement to worst", defined as the average ratio of the difference between the worst RMSE and the model being examined relative to the RMSE difference between the worst and best. The final column reports the median RMSE. The sample period is from 1926 to 2022.

4.4. Robustness

In this section, we verify the robustness of our out-of-sample results to changing our definition of how idiosyncratic variance is calculated, to limiting our sample to liquid firms (those included in the S&P 500), and to the choice of winsorization of the forecasts.

We estimate idiosyncratic variance in two alternative ways, designed to reduce the noise inherent in an estimation that uses relatively few observations (around 20 daily observations within the month) to obtain the parameters in Eq. (1). First, we estimate Eq. (1) without the HML and SMB factors, assuming a CAPM model. Second, we follow Welch (2022) and start by winsorizing daily firm returns to an interval of minus 2 to plus 4 times the contemporaneous market return and then estimate Eq. (1) using three months of data (but calculate *IVAR* using only the residuals in month t).

The results, presented in Panels A-B of Appendix A, show that the specific factor model used to calculate idiosyncratic variances does not change the conclusion that the ARMA model performs best in forecasting future *IVAR*. This is comforting, suggesting the results in this paper are not sensitive to the particular asset pricing model used to compute idiosyncratic variances.

Our next set of robustness checks attempts to ensure that microstructure noise, such as infrequent trading, does not drive our results. Given that prior literature has suggested microstructure noise may drive the relation between certain measures of idiosyncratic risk and returns (Bali and Cakici, 2008; Han and Lesmond, 2011; Bergbrant and Kassa, 2021), this is an important consideration for any paper measuring idiosyncratic volatility. Our first approach to account for this noise is to correct for autocorrelation in the daily residuals using the French et al. (1987) correction. For these models, the *IVAR* is calculated by adding 2 times the sum of the products of each daily return with the prior day's return. Our second approach is to consider a sample which only includes firms in months when they were part of the S&P 500. The results for these tests are presented in Panels C and D of Appendix A. The martingale model now performs substantially worse with the percentage of firms for which it is the best model decreasing to less than 10%; and even decreasing below 6% for the S&P 500 firms (5.77%), which are arguably the firms with the least microstructure noise. Importantly, the

ARMA(1,1) model continues to produce the best forecasts for the largest number of firms. In fact, for the most liquid (S&P 500) firms, the ARMA (1,1) produces the best forecast for over 40% of firms (42.37%).

For the main out-of-sample results in the paper, we winsorize our monthly forecasts at 0 and 0.5. For robustness, we also estimate the main results without winsorization, as well as with monthly winsorization at 2.5% and 97.5% of the cross-sectional distribution of the IVAR forecasts. The results are presented in Panels E and F of Appendix A. While the mean RMSE are much higher (extremely so for some models) without winsorization,[15] the ARMA(1,1) model remains the best model for all statistics, and has the lowest RMSE for over 35% of firms. This remains true for the alternative (2.5%, 97.5%) winsorization scheme.

5. The idiosyncratic volatility pricing puzzle

In this section, we reconsider the IVOL puzzle with different models measuring expected idiosyncratic risk (Section 5.1). We then further analyze why the martingale model delivers different results than all other models, by focusing on forecasts for which the martingale performs the worst, as well as differences between the martingale and ARMA(1,1) forecasts (Section 5.2). Finally, we link extreme martingale forecasts, which render the relationship between idiosyncratic volatility and future returns negative, to explanatory variables from the IVOL literature (Section 5.3).

5.1. Revisiting the IVOL puzzle

Traditional asset pricing models such as the CAPM and APT predict that idiosyncratic volatility can be diversified away, and therefore should not be priced in equilibrium. However, subsequent studies argue that many investors might be poorly diversified (Blume and Friend 1975; Goetzmann and Kumar, 2008) and thus might require a premium for bearing idiosyncratic risk (Levy 1978; Merton 1987; Malkiel and Xu 2002).

However, a large empirical literature documents a *negative* relation between high *realized* idiosyncratic volatility and subsequent returns (Ang et al. 2006, 2009). This negative relation is often referred to in the literature as the idiosyncratic volatility (IVOL) puzzle, under the implicit conjecture that IVOL follows a martingale process and past IVOL can be used as a proxy for expected volatility. Consequently, a voluminous literature on the determinants of the "IVOL puzzle" has developed. One strand argues that investors have a preference for lottery-like payoffs; IVOL then simply proxies for desired features of a stock, such as skewness (see, for example, Barberis and Huang, 2008; Chabi-Yo and Yang, 2009; Boyer et. al. 2010; Bali et al., 2011.) Alternative explanations for the IVOL puzzle include bid-ask bounce (Fu, 2009; Huang et al., 2010), illiquidity (Han and Lesmond, 2011), fundamental uncertainty (Johnson, 2004; Jiang, et al., 2009), a missing risk factor (Chen and Petkova, 2012), and arbitrage asymmetry (Stambaugh et al., 2015). Hou and Loh (2016) provides an excellent survey of the extant literature.

A small subsequent literature suggests that when an alternative volatility model is used, namely an EGARCH model, a positive (Fu, 2009; Brockman et al., 2022), albeit fragile (Guo et al., 2014; Bergbrant and Kassa, 2021) relation between expected idiosyncratic volatility and returns emerges at the firm level.

Given that IVOL forecasts from different models (EGARCH vs. martingale) generate opposite results, it is plausible that the relation is model specific and does not reflect the true relation between expected idiosyncratic risk and returns. The actual relation between expected idiosyncratic risk and returns should be present for an idiosyncratic risk forecast that most closely approximates the true expected idiosyncratic variance. Our analysis shows that neither the martingale model nor the EGARCH model is likely to accurately approximate true expected idiosyncratic risk, whereas the ARMA(1,1) model provides the best estimate.

In Table 5, Panel A, we use the seven proxies we developed in this paper in a univariate Fama-MacBeth (1973) regression framework to investigate the empirical relation between expected idiosyncratic risk and returns.[16] The table shows that the only proxy for expected idiosyncratic volatility that is statistically significantly related to returns is the one derived from the martingale model. While the coefficients are always negative, all the other proxies for expected idiosyncratic risk show no statistically significant relation with returns, and the *t*-stats are less than $|1|$ in all cases. The failure to reject the null of no effect also holds for the best proxy for expected idiosyncratic variance identified in this paper, i.e., the ARMA(1,1) model. Note also that the martingale model generates the lowest adjusted R-square of all models.

In Panel B of Table 5, we report multivariate tests controlling for typical predictors of the cross-section of stock returns including market beta, size, book-to-market, momentum, lagged returns (prior month's return), turnover, and the coefficient of variation of turnover in our Fama-MacBeth (1973) regressions (see Huang et al., 2010 for a similar specification). Including an extensive set of independent variables is particularly important in our context as IVAR itself is estimated using only the Fama and French 3-factors,[17] and hence could be related to returns due to its correlation with any omitted risk factor. Again, a significant relation between expected IVOL and returns only arises using the forecasts from the martingale model.

We also acknowledge the possibility that different IVAR models might work better for different firms, or at different points in time. Results in the previous sections indicate that the ARMA(1,1), although best overall, is not the best for all firms. In the second to last columns of Panel A and B of Table 5, we allow the volatility model to differ across firms by using forecasts from the model with the overall lowest out-of-sample RMSE for each firm. Although all forecasts are out-of-sample, it is important to note that this approach has a look-ahead bias in the sense that the best IVAR model for each firm is determined ex-post. This exercise further corroborates our key result, as the IVOL coefficient remains insignificant.

We also allow different IVAR models to be used in different periods and for different firms, by back testing our models each month. We do this by comparing the past out-of-sample RMSE forecasts generated by each model, for each firm, in the 10 years leading up to time t. We then use the best IVAR model to generate our realized variance expectation for the next month $t+1$. The results are presented in the last column of Table 5 (Min_RMSE), showing again an insignificant IVOL coefficient.

Overall, our results suggest that the relation between idiosyncratic volatility and returns is flat, consistent with traditional asset pricing theory, and the negative relation documented in prior literature is likely an artifact of the proxy used for expected idiosyncratic volatility rather than reflecting the true underlying relation between expected idiosyncratic volatility and returns. However, it is important to note that for our tests, we do not know the magnitude of type II errors, so we cannot unambiguously conclude that no relation between idiosyncratic volatility and returns exists. It is possible that all our proxies contain sufficient noise to mask a true relation between expected idiosyncratic risk and returns.

5.2. Why does the martingale model generate the IVOL puzzle?

Although there is a voluminous literature trying to explain the

[15] Extreme forecasts for the non-linear AR models render the average statistics meaningless. These models sometimes generate very poor and unrealistic forecasts as a result of extreme and unrealistic parameter estimates. Note that for the ARMA(1,1) model, winsorizing makes little difference.

[16] To be consistent with the IVOL literature, we use the square root of the IVAR forecast, but using the IVAR forecast itself produces analogous results.

[17] Estimating IVAR using models with more risk factors is difficult because we use daily data within the month.

Table 5
Is expected idiosyncratic volatility priced?

Panel A: Univariate					w/ Mkt				
	Martingale	ARMA(1,1)	ARNL	HAR	ARMA(1,1)	ARNL	HAR	Best_Overall	Min_RMSE
Intercept	1.014***	0.957***	0.936***	0.972***	0.943***	0.914***	0.956***	0.877***	0.949***
	(5.79)	(5.74)	(5.72)	(5.76)	(5.68)	(5.64)	(5.72)	(5.33)	(5.76)
IVOL	−2.404**	−1.113	−0.941	−1.112	−0.926	−0.731	−0.944	−0.541	−0.970
	(−2.24)	(−0.75)	(−0.65)	(−0.81)	(−0.63)	(−0.51)	(−0.69)	(−0.37)	(−0.68)
Adj R-Sq	0.020	0.026	0.024	0.023	0.025	0.024	0.023	0.025	0.024

Panel B: Multivariate					w/ Mkt				
	Martingale	ARMA(1,1)	ARNL	HAR	ARMA(1,1)	ARNL	HAR	Best_Overall	Min_RMSE
Intercept	3.094***	2.998***	3.013***	2.987***	2.961***	2.997***	2.969***	2.973***	3.048***
	(8.67)	(8.70)	(8.61)	(8.53)	(8.55)	(8.56)	(8.46)	(8.36)	(8.56)
IVOL	−1.740***	−0.996	−0.163	−0.662	−0.894	0.027	−0.472	0.439	−0.860
	(−2.64)	(−0.82)	(−0.15)	(−0.64)	(−0.76)	(0.02)	(−0.46)	(0.37)	(−0.84)
BETA	0.198	0.200	0.182	0.197	0.207	0.185	0.198	0.171	0.188
	(1.27)	(1.35)	(1.21)	(1.30)	(1.39)	(1.23)	(1.30)	(1.15)	(1.24)
LN(ME)	−0.125***	−0.122***	−0.112***	−0.118***	−0.119***	−0.110***	−0.116***	−0.100***	−0.121***
	(−4.33)	(−4.78)	(−4.28)	(−4.47)	(−4.59)	(−4.17)	(−4.33)	(−3.91)	(−4.63)
LN(BEME)	0.189***	0.185***	0.192***	0.189***	0.189***	0.194***	0.192***	0.196***	0.189***
	(3.55)	(3.49)	(3.60)	(3.56)	(3.55)	(3.62)	(3.60)	(3.68)	(3.53)
MOM	0.027**	0.028**	0.028**	0.027**	0.028**	0.028**	0.027**	0.028***	0.026**
	(2.42)	(2.55)	(2.50)	(2.37)	(2.55)	(2.49)	(2.39)	(2.61)	(2.34)
LAGRET	−0.059***	−0.061***	−0.060***	−0.061***	−0.061***	−0.060***	−0.060***	−0.061***	−0.061***
	(−13.92)	(−14.49)	(−14.45)	(−14.47)	(−14.46)	(−14.44)	(−14.45)	(−14.75)	(−14.40)
LN(TURN)	−0.165***	−0.155***	−0.171***	−0.162***	−0.158***	−0.175***	−0.166***	−0.183***	−0.154***
	(−2.92)	(−2.89)	(−3.13)	(−2.93)	(−2.91)	(−3.21)	(−3.00)	(−3.40)	(−2.85)
LN(CVTURN)	−0.347***	−0.339***	−0.363***	−0.343***	−0.336***	−0.365***	−0.345***	−0.379***	−0.354***
	(−6.01)	(−6.26)	(−6.66)	(−6.18)	(−6.11)	(−6.65)	(−6.13)	(−6.86)	(−6.15)
Adj R-Sq	0.064	0.065	0.065	0.064	0.065	0.065	0.064	0.065	0.065

This table reports cross-sectional Fama and MacBeth (1973) regression results of forecasting one month ahead (time *t*) stock returns. IVOL is the square root of the out-of-sample forecast of IVAR for time *t* based on prior data (up until time *t-1*) using 7 different forecasting models and 2 selection models. Panel A shows univariate results, while Panel B presents the multivariate version. Newey and West (1987) corrected *t*-statistics with three lags are reported in parentheses, and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels respectively. The data are from 1963:07 to 2022:12.

negative relation between martingale IVOL and returns, a fully convincing explanation remains elusive (see, for example, the review in Hou and Loh, 2016 and the references therein). In this section, we link the puzzle to the fact that the martingale model generates poor forecasts for future idiosyncratic variances. Even for firms where it produces the "best" forecasts, the martingale forecasts are generally poor with large forecast errors. So, why do forecasts produced by the martingale model generate a negative relation to future returns, while all other models, including the better ARMA(1,1) forecasts, show an insignificant relation?

To understand this, let's first assume that there is no relation between expected IVOL and returns (as suggested by all our models except for the martingale model). While there will still be some firm months with high expected IVOL and subsequently low returns, the impact of those observations would be offset by firm months with high expected IVOL and high returns, leading to an insignificant relation. The same would be true for firm months with low expected IVAR and subsequent (low/high) returns. Put differently, it should then be the case that in the data the four possible quadrants of current volatility forecasts (high vs low) and future returns (high vs low) cancel each other out.

However, outliers (observations with extreme expected variances and subsequent extreme returns) can have a large impact on the estimated coefficients and could easily cause a spurious relation (negative or positive) between expected volatility and returns, especially if the outliers are more likely to occur in one of the four quadrants.

The well-known fact that the realized variance distribution is skewed plays a key role in the martingale model generating a negative relationship between current volatility forecasts and future returns. First, this skewness renders the martingale model particularly sensitive to outliers in the right tail. In fact, we find that the worst forecast errors (both over-predictions and under-predictions) are concentrated in the high realized variance quadrants.

Second, it is well known that the contemporaneous relation between IVAR and equity returns is positive (limited liability is one source of such relationship). Thus, if the erroneous outliers over-predict future variances (and hence, returns), the relationship between current IVOL and future returns could easily turn negative, as it associates, erroneously, low future return firm months with previously high realized variance months. In contrast, current relatively high IVAR observations that severely under-predict future realized variances, are associated with high returns in the future and cannot be the source of a negative relationship between current IVOL and future returns.

Outlying under-prediction observations associated with low realized variances could also turn the relationship negative, associating, erroneously, high future returns with previously low realized variances. However, because there are few large forecast errors in this quadrant, the low current realized variance, high future returns quadrant is not likely responsible for the negative IVOL relation. Hence, it is likely the high volatility quadrant, which generates outliers potentially inducing a spurious positive or negative relation between volatility forecasts and returns.

To test the conjecture that the negative relation between the martingale forecasts and returns is driven by a few outliers, we remove observations with the worst possible forecasts and verify how the relation between idiosyncratic volatility and future returns changes. Note that if the negative relation between martingale IVAR and returns found in the data reflects investors actually "pricing" IVOL *we would expect it to be stronger when poor forecasts (ex-post) are excluded* and be robust to exclusions of subsets of firms.

In Panel A of Table 6, we first eliminate 0%−1% of firms (in 0.2% increments) each month where the martingale forecasts overestimate (underestimate) realized IVAR the most. We then report the regression coefficients on IVOL for both the univariate and multivariate cross-sectional models (the multivariate model is the same as in Table 5, but we suppress the coefficients on the controls from brevity). As expected, removing firms where the martingale model overestimates IVAR the

Table 6
Forecast errors and the IVOL puzzle.

Panel A: Drop highest Martingale IVOL forecast errors

	Overestimation of IVAR		Underestimation of IVAR	
	Univariate	Multivariate	Univariate	Multivariate
Delete top 0%	−2.404**	−1.740***	−2.404**	−1.740***
	(−2.24)	(−2.64)	(−2.24)	(−2.64)
Delete top 0.2%	−1.924*	−1.397**	−3.776***	−2.604***
	(−1.73)	(−2.03)	(−3.57)	(−4.03)
Delete top 0.4%	−1.281	−0.859	−4.496***	−2.975***
	(−1.11)	(−1.21)	(−4.33)	(−4.70)
Delete top 0.6%	−0.697	−0.356	−4.964***	−3.183***
	(−0.58)	(−0.48)	(−4.83)	(−5.06)
Delete top 0.8%	−0.207	0.152	−5.343***	−3.376***
	(−0.17)	(0.20)	(−5.26)	(−5.38)
Delete top 1.0%	0.255	0.628	−5.649***	−3.562***
	(0.20)	(0.79)	(−5.61)	(−5.71)

Panel B:Drop highest Martingale-ARMA Forecasts

	Martingale Forecast		ARMA(1,1) Forecast	
	Univariate	Multivariate	Univariate	Multivariate
Delete top 0%	−2.404**	−1.740***	−1.113	−0.996
	(−2.24)	(−2.64)	(−0.75)	(−0.82)
Delete top 1%	−2.337*	−1.855**	−0.741	−0.913
	(−1.94)	(−2.47)	(−0.49)	(−0.73)
Delete top 2%	−2.305*	−1.783**	−0.551	−0.717
	(−1.79)	(−2.21)	(−0.36)	(−0.57)
Delete top 3%	−2.322*	−1.898**	−0.452	−0.754
	(−1.74)	(−2.25)	(−0.29)	(−0.59)
Delete top 4%	−2.303*	−1.856**	−0.365	−0.647
	(−1.66)	(−2.12)	(−0.23)	(−0.50)
Delete top 5%	−2.219	−1.861**	−0.258	−0.676
	(−1.55)	(−2.13)	(−0.17)	(−0.54)
Delete top 6%	−2.087	−1.628*	−0.132	−0.526
	(−1.43)	(−1.79)	(−0.08)	(−0.41)
Delete top 7%	−2.020	−1.465	−0.066	−0.400
	(−1.35)	(−1.56)	(−0.04)	(−0.31)
Delete top 8%	−1.810	−1.269	0.058	−0.321
	(−1.19)	(−1.35)	(0.04)	(−0.25)
Delete top 9%	−1.757	−1.164	0.091	−0.261
	(−1.14)	(−1.22)	(0.06)	(−0.20)
Delete top 10%	−1.680	−1.207	0.146	−0.379
	(−1.07)	(−1.25)	(0.09)	(−0.30)

This table reports cross-sectional Fama and MacBeth (1973) regression results of forecasting one month ahead (time *t*) stock returns using the martingale IVOL forecast (Panel A) and martingale and ARMA forecasts (Panel B). In Panel A, we drop observations each month for which the martingale performs the worst in terms of overpredicting (Model 1 and 2) and underpredicting (Model 3 and 4) realized variances. While additional controls are included in Models 2 and 4, they are suppressed for brevity. In Panel B, we drop observations based on the difference between the martingale forecasts compared to the ARMA(1,1) forecasts. Again, the additional controls in Models 2 and 4 are suppressed for brevity. Newey and West (1987) corrected t-statistics with three lags are reported in parentheses, and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels respectively. The data are from 1963:07 to 2022:12.

most leads to a less negative relation (i.e., the magnitude is smaller, in absolute value), whereas removing observations where the martingale model underestimates realized IVAR the most leads to a more negative relation (because we are removing observations from the high forecast, high future return quadrant which induces a positive relation). Interestingly, eliminating just the top 0.4% of firms with the largest overestimation already renders the relationship statistically insignificant for the remaining firm-month observations. At a 1% (0.8%) cut, the relationship even turns positive for the univariate (multivariate) models. It is important to note that these cuts are made ex-post, and we would expect that the relation between expected IVOL and returns would change in the direction observed if we excluded a large number of firm months with over- or underpredictions. However, these results indicate that the relation between the martingale IVOL forecast and returns is strongly influenced by a few firm months for which the martingale model does the worst job at forecasting and the negative relation can be overturned by eliminating the observations showing the worst overprediction of realized variances.[18]

We have previously shown that the ARMA(1,1) does a stellar job in forecasting IVAR, possibly because it allows the reversal of large shocks to variances. If the overpredictions from the martingale model occur because of firms with large past shocks to their IVAR which the martingale model fails to reverse, we would expect that eliminating firms where the martingale model has the highest IVAR predictions compared to the ARMA(1,1) model would also diminish the negative relation between martingale IVAR and returns. The benefit of this, compared to the prior results, is that the exclusions are made ex-ante (based solely on differences in expectations). Naturally, we would need to exclude more firm months as we do not know which observations produce the greatest outliers. The results are presented in Panel B of Table 6 (left column). A similar pattern as before (in Panel A) is present when dropping these firm months. Dropping 5% (7%) of the firms each month for which the martingale fails to reverse large past IVAR (i.e. the martingale generates the highest forecast compared to the ARMA model) in the univariate (multivariate) regressions renders the relation insignificant. Note that the univariate regressions feature more negative coefficients and require fewer observations to be dropped to render the relationship insignificant.. Importantly, dropping the same firm months, the inference regarding the relationship between idiosyncratic volatility and returns does not change when ARMA(1,1) forecasts are used to measure idiosyncratic risk. (See right-hand side columns.)

5.3. *Martingale forecasts and firm characteristics*

Thus far, we have shown that the IVOL puzzle only obtains for the martingale forecasts, and that it is heavily influenced by observations for which the martingale model generates extremely poor, excessively high forecasts. For these outliers to drive the IVOL puzzle, purported explanations of the puzzle should be related to these extreme martingale forecast errors (outliers). While no article has yet claimed to fully account for the IVOL puzzle, Hou and Loh (2016) provide a comprehensive examination of various explanations and attempt to quantify how much of the puzzle they explain. They organize these explanations into three groups: lottery preferences of investors (measured by skewness, coskewness, expected idiosyncratic skewness, maximum daily return, and retail trading proportion); market frictions (measured by one-month return reversal, a variable that is already included in our standard risk controls, the Amihud illiquidity measure, zero return proportion, and bid-ask spreads) and a third group that includes all other variables.[19] In addition, we investigate variables that mimic the size screens that were shown to be important in Bali and Cakici (2008) (namely, dummies indicating stocks with prices lower than $5 and stocks with a market capitalization lower than the 5th percentile of the NYSE stocks ranked on market capitalization). All the variables are defined in the data section.

Because the IVOL puzzle is driven by the observations for which the martingale model overestimates IVOL the most, we conjecture that candidate explanations for the IVOL puzzle more correlated with extreme martingale overestimations (compared to underestimations) should show stronger explanatory power of the puzzle. Such relationship is not likely captured well by linear regressions (recall that just dropping 0.4% of the extreme forecast error observations suffices to render the martingale IVOL-return relationship insignificant). We

[18] Note that our winsorizing of forecasts mitigates the outlier problem, but we observe the exact same phenomenon using the raw, non-winsorized forecasts.

[19] We exclude the variables in the last category as they substantially reduce our sample size, but confirm in unreported tests that including them has a negligible impact on the explanatory power of the models.

therefore resort to several regression methodologies that measure relationships in the tail: a linear probability (LinProb) model where we create a dummy for the top decile of martingale over-predictions (i.e. the negative of the martingale forecast errors); a quantile regression (QREG) focusing on the top 10% over-predictions, and a weighed least squares (WLS) regression where the weights are proportional to the square root of the percentile rank of the (negative of the) forecast error. As independent variables, we use the usual risk controls that we employed in the return model (Model 1) and then add the various variables proposed in Hou and Loh (2016) in Model 2. When we add the size related variables from Bali and Cakici (2008), we drop the size risk control to avoid multi-collinearity (See Internet Appendix 4, Table 3).

In Table 7, for brevity, we only show the coefficients from the multivariate results for the WLS regressions (the coefficients from the other two models, the linear probability model and the quantile regression models, are relegated to Internet Appendix 4, Table 1). In these models, all continuous control variables are winsorized (at 0.5% and 99.5%) and all continuous variables are standardized with a mean of 0 and a standard deviation of 1 to ease interpretation. Standard errors are clustered at both the firm and time (month) level (i.e. 2-way clustered standard errors). We also report a covariance decomposition, indicating the ratio of the covariance between the fitted value and the coefficient times the independent variable over the variance of the fitted value. These covariance contributions add to 100%, and use the average across the 3 models discussed above (WLS, linear probability, and quantile regression). Similarly, the "rank average" reports the relative importance (in terms of the absolute magnitude of the coefficient) of the variable in question across the three different specifications. We rank the independent variables based on the magnitude of the absolute coefficients from largest (1) to smallest (total number of independent variables) and average across the specifications.

We first focus on the multivariate regressions that exclude the maximum return variable (MAX) from Bali et al. (2011). According to Hou and Loh (2016), this variable is very highly correlated with IVOL and it is hard to see it as a proper measure of skewness or lottery preferences. The inclusion of MAX also has a huge impact on the other explanatory variables, as including it changes the sign of various coefficients compared to when the base model 1 is augmented with one additional variable at a time (presented in Internet Appendix 4, Table 2A). In contrast, no coefficients change sign from the specification where only that one variable was added to the risk controls compared to Model 2, where all of them except for MAX are included. The sign consistency is also largely true when using the other econometric methodologies (not reported).

For the multivariate regressions using only standard control variables (Model 1), the economically most important coefficients (given standardization, this is measured immediately from the magnitude of the coefficients) are size (especially when the alternative size metrics are used in Internet Appendix 4, Table 3A), reversal (as measured by LAGRET) and liquidity. This is corroborated by the covariance decomposition, which shows that these variables rank as the most important across the models, and account for 36% (LNME), 27% (CVTURN), and 23% (LAGRET) of the variation respectively. When the additional control variables from Hou and Loh (2016) are added (Model 2), the results

Table 7
Negative martingale forecast error.

	(1)			(2)			(3)		
	WLS	Decompose Avg	Rank Avg	WLS	Decompose Avg	Rank Avg	WLS	Decompose Avg	Rank Avg
BETA	−0.000	1.21%	7.00	0.001	0.48%	12.33	−0.007***	−0.83%	7.00
	(−0.037)			(1.520)			(−19.651)		
LN(ME)	−0.034***	36.43%	1.00	−0.004***	6.76%	8.00	0.019***	−1.56%	4.67
	(−21.898)			(−3.011)			(22.043)		
LN(BEME)	−0.006***	0.30%	6.00	0.001	−0.04%	11.67	0.006***	0.03%	10.00
	(−6.972)			(1.211)			(14.346)		
MOM	−0.017***	5.77%	5.00	−0.009***	1.08%	8.67	−0.000	−0.03%	12.00
	(−10.416)			(−7.578)			(−0.522)		
LAGRET	0.031***	23.11%	2.33	0.022***	6.62%	4.00	−0.002**	−0.47%	9.00
	(18.889)			(14.690)			(−2.022)		
LN(TURN)	0.028***	6.59%	3.33	0.015***	0.95%	6.67	−0.009***	−0.30%	6.00
	(20.344)			(16.376)			(−13.481)		
LN(CVTURN)	0.026***	26.58%	3.33	0.023***	8.49%	4.67	0.003***	0.62%	10.33
	(23.724)			(29.701)			(6.423)		
SKEW				0.022***	5.15%	6.33	−0.043***	−6.69%	2.00
				(19.517)			(−61.189)		
COSKEW				0.003***	0.28%	10.67	−0.001***	−0.03%	11.33
				(3.948)			(−2.911)		
EXPIDIOSKEW				0.007***	2.47%	8.00	−0.000	−0.09%	13.00
				(5.392)			(−0.058)		
AMILL				0.011***	4.94%	6.67	0.001	0.94%	6.00
				(9.921)			(1.372)		
ZERORET				−0.065***	−1.81%	2.33	−0.011***	0.08%	6.00
				(−28.276)			(−9.014)		
BIDASK				0.077***	64.64%	1.00	−0.004***	−2.79%	6.67
				(36.247)			(−3.986)		
MAX							0.147***	111.11%	1.00
							(84.446)		
Adj R-Sq	0.085			0.174			0.375		

The table reports coefficients from regressing standardized martingale (negative) forecast error as the dependent variable on standardized and winsorized (at 0.5% and 99.5%) continuous independent variables using weighted least squares. All dependent and independent variables have been multiplied by the square root of the martingale error percentile rank. "Decompose Avg" is the average of the decomposed covariance using $cov\left(\widehat{y}_{i,t},\ \beta_j X_{j,i,t}\right)/var\left(\widehat{y}_{i,t}\right)$ where $\widehat{y}_{i,t}$ is the predicted y for firms i in month t and X_j is a vector of independent variables used in the model, across the WLS model, a Linear Probability model, and a Quantile Regression model (where the latter two focus on extreme decile). "Rank Avg" is the average rank (highest=1) of the magnitude of the absolute coefficient in each model averaged across the 3 models. Clustered standard errors by time and firm are reported in parentheses, and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels respectively.

Table 8
Difference between martingale and ARMA(1,1) Forecasts.

	(1)			(2)			(3)		
	WLS	Decompose Avg	Rank Avg	WLS	Decompose Avg	Rank Avg	WLS	Decompose Avg	Rank Avg
BETA	−0.016***	0.18%	7.00	−0.011***	−0.04%	10.33	−0.064***	−1.02%	7.00
	(−4.000)			(−3.109)			(−22.110)		
LN(ME)	−0.181***	40.78%	1.00	−0.001	5.30%	9.67	0.123***	−2.34%	6.67
	(−17.827)			(−0.056)			(20.702)		
LN(BEME)	−0.045***	0.35%	6.00	0.006	0.01%	12.00	0.038***	0.06%	9.00
	(−7.743)			(1.138)			(12.473)		
MOM	−0.167***	22.72%	2.00	−0.096***	5.03%	4.00	−0.032***	1.09%	8.00
	(−14.974)			(−11.132)			(−6.849)		
LAGRET	0.105***	8.78%	4.67	0.059***	1.53%	6.67	−0.071***	−1.37%	4.33
	(8.446)			(4.989)			(−12.505)		
LN(TURN)	0.151***	5.05%	3.67	0.078***	0.73%	8.33	−0.101***	−0.37%	4.67
	(15.262)			(11.279)			(−21.825)		
LN(CVTURN)	0.156***	22.14%	3.67	0.148***	6.44%	5.00	−0.002	−0.71%	11.00
	(19.360)			(24.999)			(−0.490)		
SKEW				0.124***	4.82%	5.00	−0.265***	−6.24%	2.00
				(18.069)			(−48.579)		
COSKEW				0.013***	0.27%	9.33	−0.010***	−0.05%	13.67
				(3.197)			(−3.550)		
EXPIDIOSKEW				0.066***	1.34%	8.67	−0.014**	−0.95%	10.00
				(5.783)			(−1.993)		
AMILL				−0.022***	−2.43%	9.00	−0.101***	−2.58%	6.67
				(−2.618)			(−13.632)		
ZERORET				−0.363***	−3.38%	2.00	−0.031***	0.04%	10.00
				(−24.211)			(−5.242)		
BIDASK				0.430***	80.38%	1.00	−0.047***	−1.58%	11.00
				(34.435)			(−5.885)		
MAX							0.867***	116.01%	1.00
							(72.953)		
Adj R-Sq	0.107			0.241			0.567		

The table reports results from regressing the standardized difference between the martingale and ARMA(1,1) forecasts of IVAR as the dependent variable on standardized and winsorized (at 0.5% and 99.5%) continuous independent variables using weighted least squares. All dependent and independent variables have been multiplied by the square root of the martingale error percentile rank. "Decompose Avg" is the average of the decomposed covariance using $cov\left(\widehat{y}_{i,t},\ \beta_j X_{j,i,t}\right) / var\left(\widehat{y}_{i,t}\right)$ where $\widehat{y}_{i,t}$ is the predicted y for firms i in month t and X_j is a vector of independent variables used in the model, across the WLS model, a Linear Probability model, and a Quantile Regression model (where the latter two focus on the 90th percentile). "Rank Avg" is the average rank (highest=1) of the magnitude of the absolute coefficient in each model averaged across the 3 models. Clustered standard errors by time and firm are reported in parentheses, and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels respectively.

continue to suggest a large role for liquidity, with zero returns and bid-ask spreads being most important (as judged by the magnitude of the coefficients). The coefficients for the important variables range between 0.02 and 0.08, and we observe similar coefficients for the linear probability model (Internet Appendix 4, Table 1).

The MAX return variable is considered separately (Model 3) because it has a massive effect on the probability of observing a large forecast error. Its coefficient is of the order of 0.15 and its inclusion in the regression changes the sign of the coefficients on many variables, while also reducing them to smallish numbers. The R^2 more than doubles. One exception is the skewness coefficient which now becomes negative, likely driven by its high correlation with the MAX variable which is an indirect indicator of the potential to positive skewness and reflects a realization of high skewness. In sum, the MAX return variable is a good proxy for martingale overestimations of variances. Internet Appendix 4, Table 1 shows that alternative estimation approaches (including the linear probability model, and quantile regressions) lead to similar conclusions.[20] For the quantile regressions, the key variable, when excluding the maximum return variable, is, in a more dominant fashion than seen using other approaches, the bid-ask spread. Small, illiquid firms, which are prone to some extreme return behavior in particular firm months, appear to be the drivers of the large forecast errors for the martingale model and the IVOL puzzle.

We perform the same analysis for the difference between the martingale and ARMA(1,1) forecasts. The results, using the WLS regressions, are reported in Table 8 (as well as Internet Appendix 4). Perhaps not surprisingly, we find that the same variables are associated with larger differences between the two forecasts. In particular, the most important variable is again the MAX variable, which renders other variables much less economically important or even changes the sign of the coefficients (e.g. for the bid-ask spread). When the MAX variable is not included in the regression, the most important variables, economically, are the bid-ask spread and zero returns (the latter with a negative sign). Size indicators also play an important role, especially when measured by the alternative size dummies (Internet Appendix 4, Table 3B).

6. The cross-section of expected idiosyncratic volatility

Given that the ARMA (1,1) model is clearly the best model, we analyze what drives the ARMA(1,1) parameters and use the model forecasts to characterize the cross-sectional and time series behavior of expected idiosyncratic volatility for US firms.

6.1. ARMA(1,1) expected idiosyncratic volatility

We first provide further insights on how the ARMA model produces

[20] While the standard errors for the WLS and Linear Probability Models use two-way clustered errors, the Quantile regression models use robust standard errors. For the latter, the standard errors are likely mis-specified, and we only rely on that model for interpreting the magnitude of the coefficients and for covariance decomposition.

expected idiosyncratic variances from past information, in particular past realized variances through the autoregressive parameter, and past variance shocks through the moving average coefficient. Then we investigate how expected idiosyncratic volatility varies with firm characteristics.

Models 1 and 2 of Table 9 reports coefficients from a panel regression, regressing the estimated autoregressive and moving average parameters on firm characteristics. Because the parameters result from our out-of-sample analysis with rolling estimations, we have a panel of such coefficients. To facilitate interpretation of the results, all independent variables are standardized to have zero mean and standard deviation of 1.

Focusing on the AR-coefficients (Model 1), because all independent variables are standardized, the constant in the regression represents the average autoregressive coefficient, which is 0.63. This is quite very far away from the implied value of 1 imposed by the martingale model, and similarly far away from the average persistence coefficient in an AR(1) model, which is only 0.27 (not reported). The largest economic effects arise from the size and the book-to-market variables, with firms one standard deviation larger than the average featuring a 0.04 higher autoregressive coefficient, firms with book-to-market values one standard deviation above the mean a 0.05 higher coefficient. These effects are highly statistically significant. The turnover variables (level and coefficient of variation) also generate statistically significant effects with higher turnover (variation) lowering the AR coefficients. All other effects are economically very small.

We also report similar results for the moving average coefficients in Model 2. The average moving average coefficient is 0.41, indicating that the best forecast typically involves reversing large shocks to realized idiosyncratic variances. This explains why economically neither a martingale model, nor a simple autoregressive model can fit the data well. When shocks are small, an autoregressive model with a coefficient of around 0.63 works well. When a large positive shock to the realized variance happens, the optimal ARMA forecast lowers the effective persistence considerably; when a negative shock happens, the forecast increases the effective persistence of the process. The importance of the MA coefficient can be gleaned from Fig. 1, where we rank firms in deciles based on their change in IVAR between time *t-1* and *t*. We then show the average IVAR in subsequent time periods for the most extreme deciles. The blue line shows the average IVAR for firms with the largest increases in IVAR from *t-1* to *t* (the graph shows this to represent a shift from about 0.04 to 0.14), and the red line shows the average IVAR for the decile with the largest decreases in IVAR between *t-1* and *t* (representing roughly the opposite switch). Clearly, we see strong mean-reversion in the IVAR levels, which the ARMA model accomplishes through the MA-coefficient.

When looking at the impact on the MA coefficient, size and book-to-market again generate the largest positive coefficients, with large value firms having higher moving average coefficients (0.036, respectively 0.038 per standard deviation away from the mean). The only other economically important effect is recorded for the coefficient of variation of turnover, with a −0.036 coefficient.

Fig. 2 provides more intuition on the relative performance of the ARMA(1,1) model relative to simpler but nested models such as the martingale and the AR(1) model. We split the firms every month (at time *t-1*) based on the change in IVAR from the previous month (from *t-2* to *t-*

Table 9
What drives variation in the ARMA(1,1) parameters and IVOL.

	(1) AR	(2) MA	(3) ARMA(1,1) IVOL
Intercept	0.625***	0.412***	0.481***
	(173.820)	(132.932)	(133.344)
BETA	0.004	−0.002	0.022***
	(1.478)	(−0.954)	(13.002)
LN(ME)	0.041***	0.036***	−0.159***
	(9.293)	(8.910)	(−40.949)
LN(BEME)	0.050***	0.038***	−0.034***
	(19.969)	(15.656)	(−18.808)
MOM	−0.004***	−0.001	−0.018***
	(−2.877)	(−0.428)	(−6.808)
LAGRET	0.002*	0.001	0.009***
	(1.685)	(0.927)	(4.036)
LN(TURN)	−0.018***	−0.010***	0.097***
	(−5.573)	(−3.204)	(41.856)
LN(CVTURN)	−0.037***	−0.036***	0.071***
	(−12.544)	(−12.970)	(34.314)

The table shows a panel regression of the AR (MA) coefficient as well as the annualized volatility forecast (SQRT(Monthly VAR Forecast × 12)) of the ARMA (1,1) model on firm characteristics. All firm level explanatory variables have been winsorized at the 0.5% and 99.5% levels and all independent variables have been standardized with a mean of 0 and standard deviation of 1. Clustered standard errors by time and firm are reported in parentheses, and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels respectively.

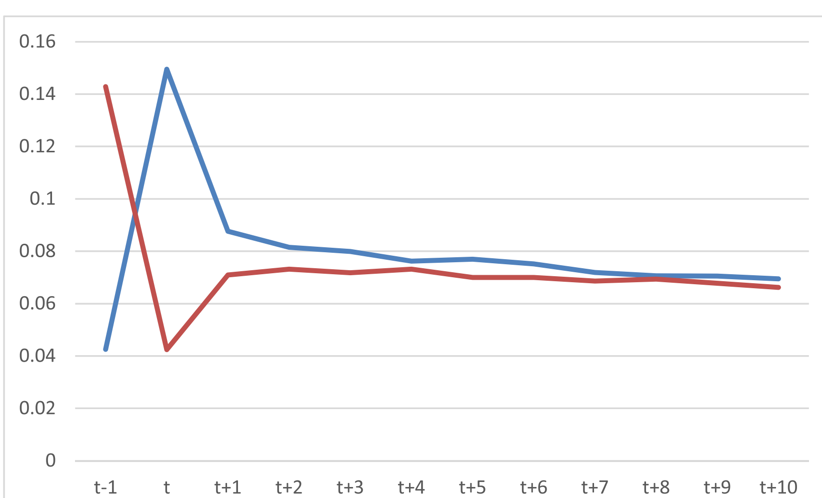

Fig. 1. This figure shows the firms with the largest changes in IVAR from time *t-1* to time *t* organized in deciles, and the subsequent mean IVAR for the two extreme deciles. The blue line shows the average IVAR for firms with the largest increases in IVAR from *t-1* to *t*, and the red line shows the average IVAR for the decile with the largest decreases in IVAR between *t-1* and *t*.

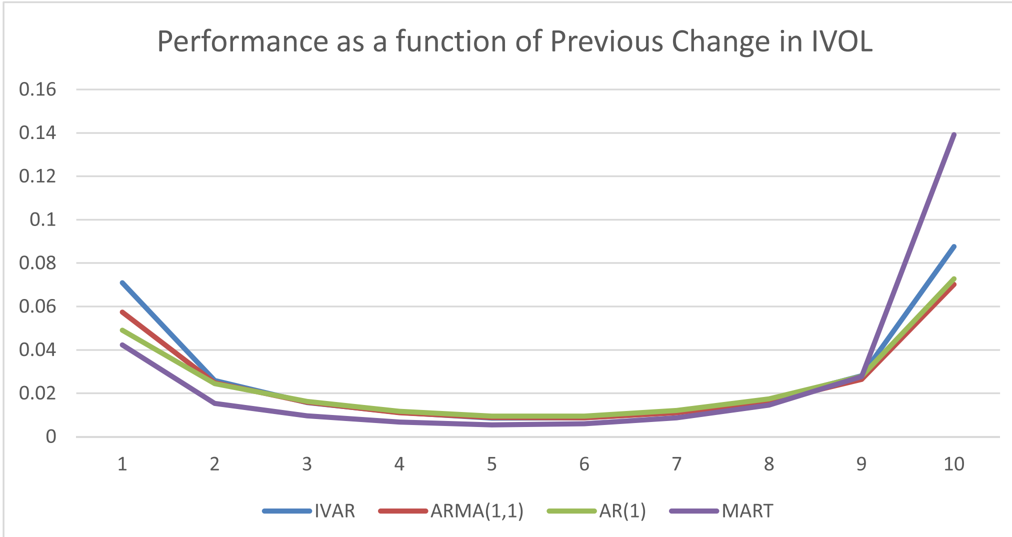

Fig. 2. This figure shows the mean IVAR forecast, as well as actual IVAR, for month *t* where the forecasts are based on data up until *t-1*. The forecasts are generated separately for each decile of change in the forecast from time *t-2* to time *t-1*. Hence, the lowest decile (1) shows the forecasts for the firms with the lowest (most negative) change in IVAR from time *t-2* to time *t-1*, and the highest decile (10 - to the right on the graph), shows the forecasts for the decile with the largest changes in IVAR from time *t-2* to time *t-1*.

1) and organize them into 10 deciles. Hence, the firms in the first decile just experienced large negative shocks to the realized variance, whereas the firms in the 10^{th} decile experienced large positive shocks to the realized variance. The figure plots the mean realized variance (IVAR) at time *t* and the average forecast, based on data up until *t-1*, for each of the 10 deciles for the three aforementioned models. Thus, in the figure on the left, the lowest decile (indicated by 1) shows the forecasts for the firms with the lowest (most negative) change in IVAR from time *t-2* to time *t-1*, and on the right, shows the forecasts for the decile with the largest changes in IVAR from time *t-2* to time *t-1*, the highest decile (indicated by 10).

As expected, the martingale model overpredicts IVAR for firms with recent large jumps in IVAR and underpredicts the IVAR for firms with recent large declines in IVAR (as the martingale model assumes that any change to IVAR is permanent). The predicted IVAR for firms with small changes to IVAR generally appears too low for this model, perhaps because it misses the skewness in variance. In contrast, the ARMA(1,1) model performs better at predicting IVAR after both positive and negative large changes in IVAR, suggesting that such shocks are partially reversed. Interestingly, the ARMA(1,1) model also produces better forecasts than the martingale model for firms with modest changes in IVAR. In comparison to the AR(1) model, its performance is most visibly better when there are large negative shocks to IVAR, and it is important to revise the predicted realized variance upwards which the AR(1) model fails to do. Because the AR(1) model typically imbues less persistence to the model than does the ARMA(1,1) model, it still performs well in cases where the realized variance is temporarily high.

Next, we characterize how expected idiosyncratic volatility varies with firm characteristics more directly and report the results in Model 3 of Table 9. We use the same panel regression framework as above, but the dependent variable now is the expected idiosyncratic volatility (in annualized terms) for each firm *i*, at time *t* using the ARMA(1,1) model. The average idiosyncratic volatility is around 48%. Most firm characteristics generate statistically significant deviations from the average, but few generate effects above 5% in absolute magnitude. The economically important effects are size and turnover, with firms one standard deviation above the average size (turnover) featuring 15.9% (9.7%) lower (higher) idiosyncratic volatility.

6.2. *Time series dynamics of expected idiosyncratic volatility*

In Fig. 3, we plot the time series behavior of idiosyncratic volatility over the full sample period. Panel B shows the median, 25% and 75% percentile of annualized expected idiosyncratic volatility using the ARMA(1,1) model, while Panel A shows the realized volatility for comparison. The 90% range, i.e. the 5th to 95th percentile, for median realized volatility over this long time period is [17.3%, 50.9%]. It is apparent that idiosyncratic volatility was extremely high in the early part of our sample (the 1930s) and then declined to much lower levels by 1960. However, the median idiosyncratic volatility then increases again, showing several peaks reaching levels close to those observed in the late thirties. The most recent such peak in our sample occurred during the beginning of the Covid pandemic, which is also characterized by extreme dispersion in idiosyncratic volatility across the 25% and 75% percentiles. Given the very large literature on an alleged downward trend in aggregate idiosyncratic volatility (see e.g., Campbell et al., 2001), this may be surprising. The time series pattern seems more consistent with a stationary process with occasional regime switches into high volatility periods (see also Bekaert et al., 2012). The expected idiosyncratic volatility series are obviously much smoother than the realized volatility graphs but follow the same pattern.

7. Summary and conclusion

Evaluating the riskiness of a stock portfolio requires an estimate of its expected idiosyncratic variance (IVAR). Very little systematic research exists on how to construct such estimates. In this article, we compare the performance of a large suite of different models to forecast idiosyncratic realized variances. While the popular martingale model is only best for 13.83% of firms in out-of-sample forecasts, the ARMA(1,1) model performs the best for 34.28% of firms (and if we include the ARMA(1,1) model incorporating a market variance term, it is best for 46.52% of

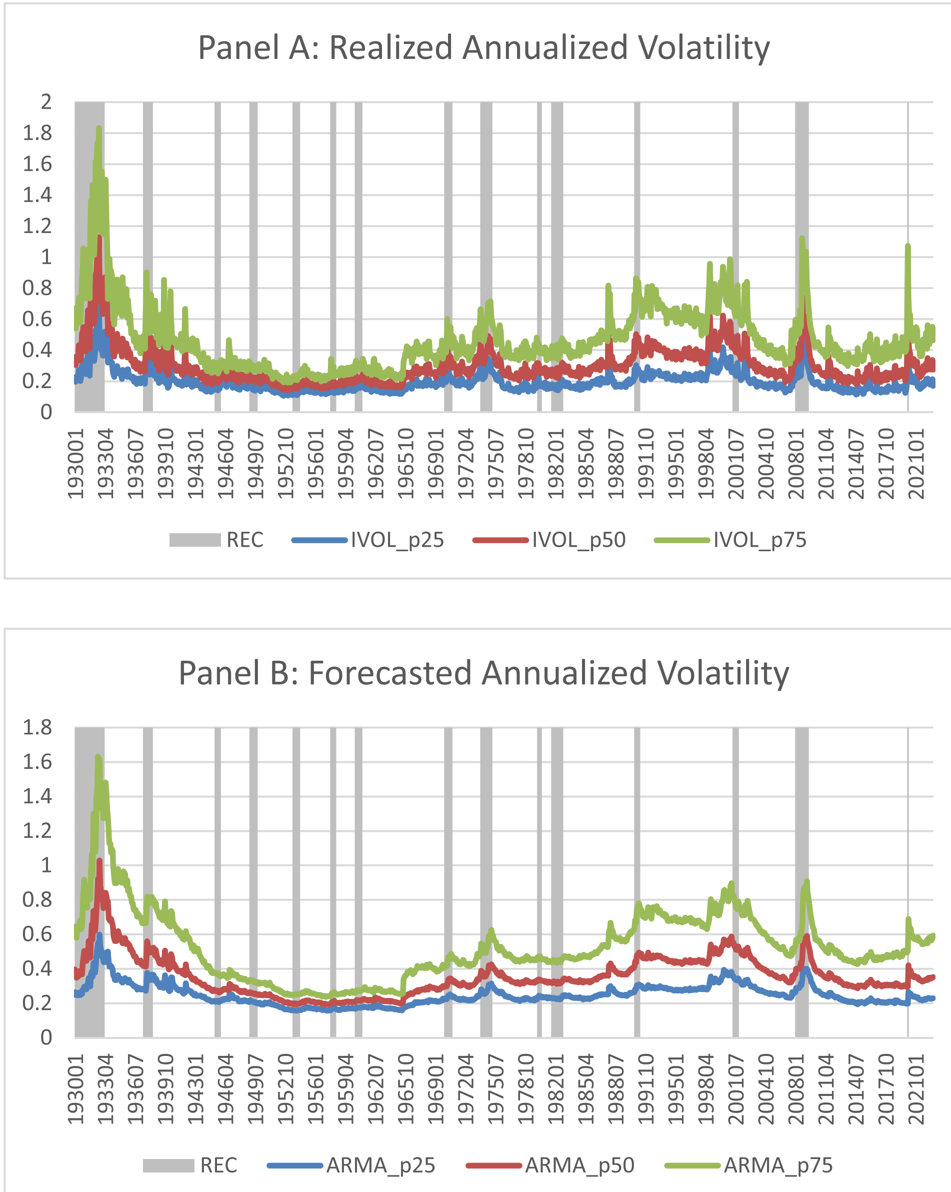

Fig. 3. Realized annual volatility.
Figure A shows the 25th, 50th and 75th percentile of realized (annualized) volatility [SQRT(Monthly IVAR × 12)] by month. Figure B shows the same percentiles of the ARMA(1,1) firm volatility forecasts.

firms). Moreover, the ARMA(1,1) model delivers lower average RMSEs than all other models in a statistically significant fashion, and also performs quite well when it is not the best model. The ARMA model produces different predictions from both a martingale model and an AR (1) model, as for a typical firm it generates persistent forecasts (with the autoregressive coefficient around 0.63, much lower than for a martingale, but much higher than observed for an AR(1) model) only when there are no large shocks to realized variance. When there are large shocks to realized variance, the forecast is adjusted in the opposite direction through a large MA coefficient (of about 0.41).

We use our various proxies for expected idiosyncratic risk to revisit the relationship between idiosyncratic risk and returns. We find that the only proxy for which the Ang et al. (2006) result (a negative relation between expected idiosyncratic risk and returns) holds, is the martingale model. For all other models, including the best ARMA(1,1) model, and when using the best IVAR model for each firm in each month, there is no relation between expected idiosyncratic risk and returns. We show that the difference is driven by a very small percentage of firm months for which the martingale model generates poor, excessively high forecasts. That is, the model fails to reverse large shocks to IVAR and therefore overestimates IVAR for firms with subsequently low returns. Eliminating firm months with high differences between the martingale and ARMA(1, 1) forecasts also renders the relationship insignificant. Not surprisingly, we find that proposed explanations for the IVOL puzzle, such as small size, skewness and poor liquidity are strongly associated with firm months featuring poor martingale forecasts, with the strongest effect associated with high bid-ask spreads. When the max return variable from Bali et al. (2011) is included, it dominates the regressions. Its explanatory power for the IVOL puzzle is highly associated with high maximum return months coinciding with firm months where the martingale model generates poor, excessively high volatility forecasts.

Our idiosyncratic volatility measures may be useful in several alternative literatures, including option pricing for individual stock options and asset management, because expected idiosyncratic volatility is necessary to determine optimal active positions. In addition, there is little consensus in the macroeconomic literature on how to measure uncertainty. Our firm-specific measure of expected idiosyncratic variance provides a viable alternative to alternative stock market variance measures that have been used in the literature, including the aggregate market volatility (see Bloom, 2009), or cross-sectional stock market dispersion (Bloom, 2009; Christiano et al., 2014).

CRediT authorship contribution statement

Geert Bekaert: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mikael Bergbrant:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Haimanot Kassa:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

None.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jfineco.2025.104023.

Appendix A. Out-of-sample Forecast Performance Robustness Checks

This table presents robustness tests where the IVAR forecasted is that from a within month CAPM Model (Panel A), 3-month FF (Panel B), autocorrelation corrected IVAR (Panel C) and subset to only firms during which time they were included in the S&P 500 (Panel D). This table shows out-of-sample statistics for the 10 models described in the data section, where the statistics presented are based on firm averages. To calculate the out-of-sample statistics, each model is fit every month for each firm to generate a forecast for the following month, using up to 10 years of data (when available) prior to the time of the forecast. Before using these forecasts to calculate the out-of-sample statistic for each firm, all monthly forecasts are winsorized at 0 and 0.5. Panel E and F present the main results (from Table 1) with forecasts winsorized at 2.5% and 97.5% (Panel E) or not at all (Panel F). In addition to the mean statistic, the table also shows the proportion of firms for which a model generates the best forecasts, as well as the average rank of the model. We also report the "improvement to worst", defined as the ratio of the difference between the worst RMSE and the model being examined relative to the RMSE difference between the worst and best, as well as the median. The sample period is from 1926 to 2022.

Panel A: **CAPM**					
Model	Mean	Best	Rank	Imp. to Worst	Median
Martingale	0.1154	13.57%	5.31	29.72%	0.0374
ARMA(1,1)	0.1111	33.89%	2.63	81.70%	0.0318
ARNL	0.1128	17.63%	3.37	70.26%	0.0326
HAR	0.1144	8.17%	4.36	56.87%	0.0347
ARMA(1,1) w Mkt	0.1124	12.47%	3.47	71.81%	0.0331
ARNL w Mkt	0.1136	9.05%	3.95	63.72%	0.0339
HAR w Mkt	0.1153	5.09%	4.92	49.80%	0.0360
Panel B: **Fama-French 3 Factors with 3-month estimation of parameters**					
Model	Mean	Best	Rank	Imp. to Worst	Median
Martingale	0.1258	13.96%	5.28	30.12%	0.0393
ARMA(1,1)	0.1214	33.21%	2.67	81.06%	0.0335
ARNL	0.1230	17.29%	3.41	69.66%	0.0342
HAR	0.1248	8.47%	4.36	56.70%	0.0365
ARMA(1,1) w Mkt	0.1226	12.51%	3.46	71.95%	0.0348
ARNL w Mkt	0.1238	9.38%	3.93	63.76%	0.0353
HAR w Mkt	0.1256	4.97%	4.88	50.22%	0.0378
Panel C: **Autocorrelation Corrected**					
Model	Mean	Best	Rank	Imp. to Worst	Median
Martingale	0.0989	9.16%	5.84	20.64%	0.0390
ARMA(1,1)	0.0914	29.73%	2.80	82.74%	0.0323
ARNL	0.0924	20.10%	3.25	76.26%	0.0329
HAR	0.0936	12.25%	3.95	68.61%	0.0339
ARMA(1,1) w Mkt	0.0928	12.36%	3.62	74.71%	0.0336
ARNL w Mkt	0.0934	10.33%	3.91	69.73%	0.0340
HAR w Mkt	0.0947	5.95%	4.63	61.67%	0.0350
Panel D: **S&P 500 Firms**					
Model	Mean	Best	Rank	Improvement to Worst	Median
Martingale	0.0124	5.77%	6.01	17.89%	0.008
ARMA(1,1)	0.0105	42.37%	2.28	89.85%	0.006
ARNL	0.0109	15.44%	3.31	76.37%	0.006
HAR	0.0117	6.69%	4.39	64.25%	0.007
ARMA(1,1) w Mkt	0.0108	16.41%	3.13	79.72%	0.006
ARNL w Mkt	0.0111	8.86%	3.86	69.57%	0.007

(*continued on next page*)

(*continued*)

HAR w Mkt	0.0120	4.46%	5.03	55.00%	0.007

Panel E: **Winsorize at 2.5% and 97.5%**

Model	Mean	Best	Rank	Imp. to Worst	Median
Martingale	0.1015	16.66%	4.94	36.01%	0.0304
ARMA(1,1)	0.0996	34.17%	2.65	79.50%	0.0272
ARNL	0.1007	16.03%	3.44	66.98%	0.0279
HAR	0.1016	8.27%	4.29	55.36%	0.0291
ARMA(1,1) w Mkt	0.1011	11.36%	3.60	66.78%	0.0286
ARNL w Mkt	0.1017	8.19%	4.13	57.78%	0.0291
HAR w Mkt	0.1028	4.60%	4.97	45.23%	0.0303

Panel F: **No Winsorization**

Model	Mean	Best	Rank	Improvement to Worst	Median
Martingale	0.1181	11.71%	5.38	31.06%	0.0325
ARMA(1,1)	0.1041	35.85%	2.47	86.23%	0.0275
ARNL	1.01E29	19.95%	3.20	74.41%	0.0283
HAR	0.3242	7.40%	4.48	56.25%	0.0306
ARMA(1,1) w Mkt	0.1117	11.89%	3.44	74.90%	0.0290
ARNL w Mkt	1.04E25	9.43%	3.91	65.97%	0.0297
HAR w Mkt	0.3571	3.78%	5.13	47.16%	0.0320

Data availability

Expected Idiosycratic Volatility (Reference data) (Mendeley Data)

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